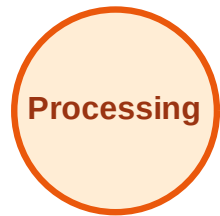


BFS Traversal

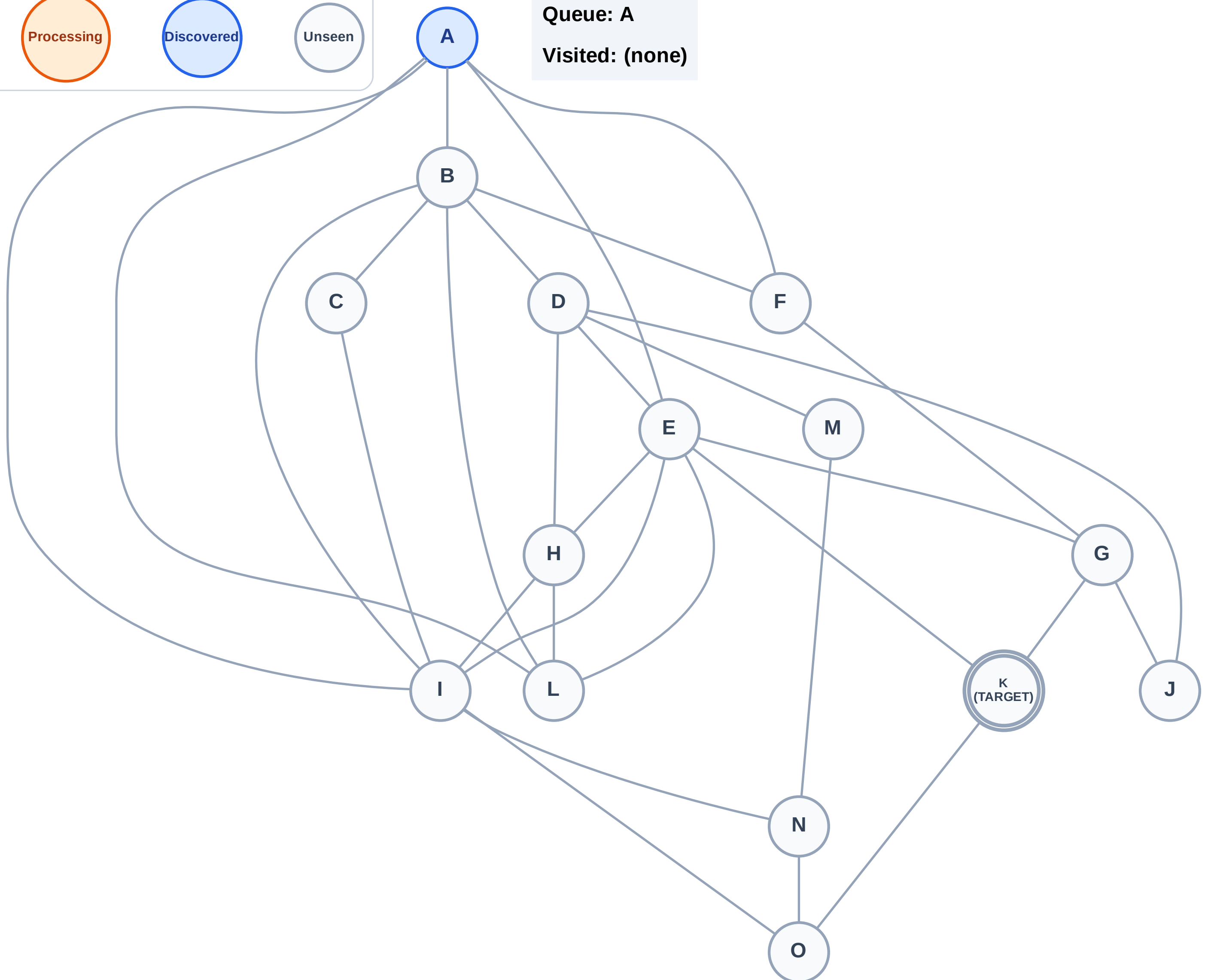
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

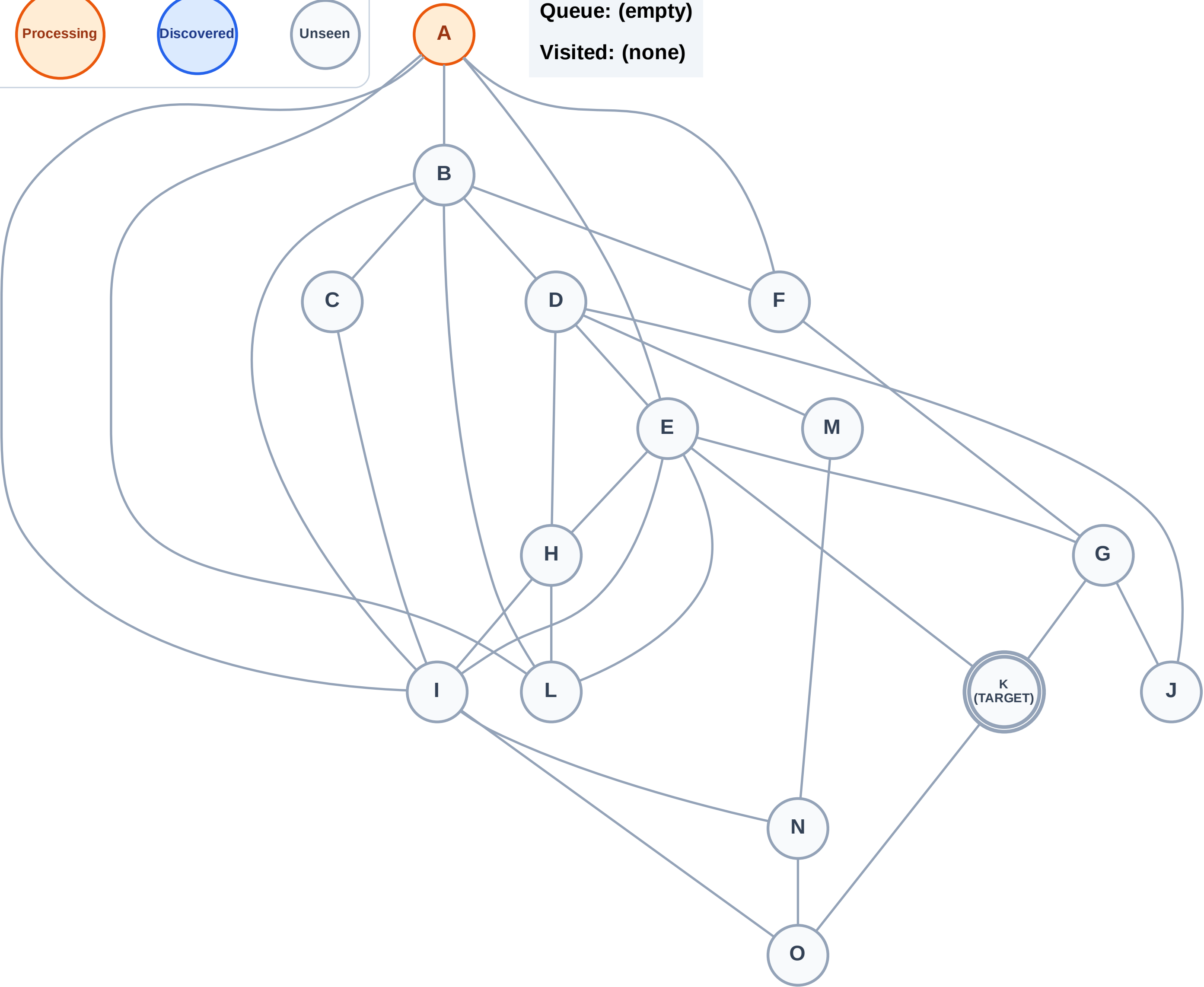
Processing

Discovered

Unseen

Queue: (empty)

Visited: (none)



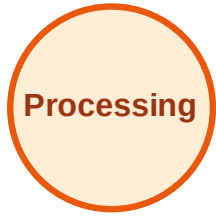
BFS Traversal

Step 3: Discover B, E, F, I, L from A

Legend



Complete



Processing



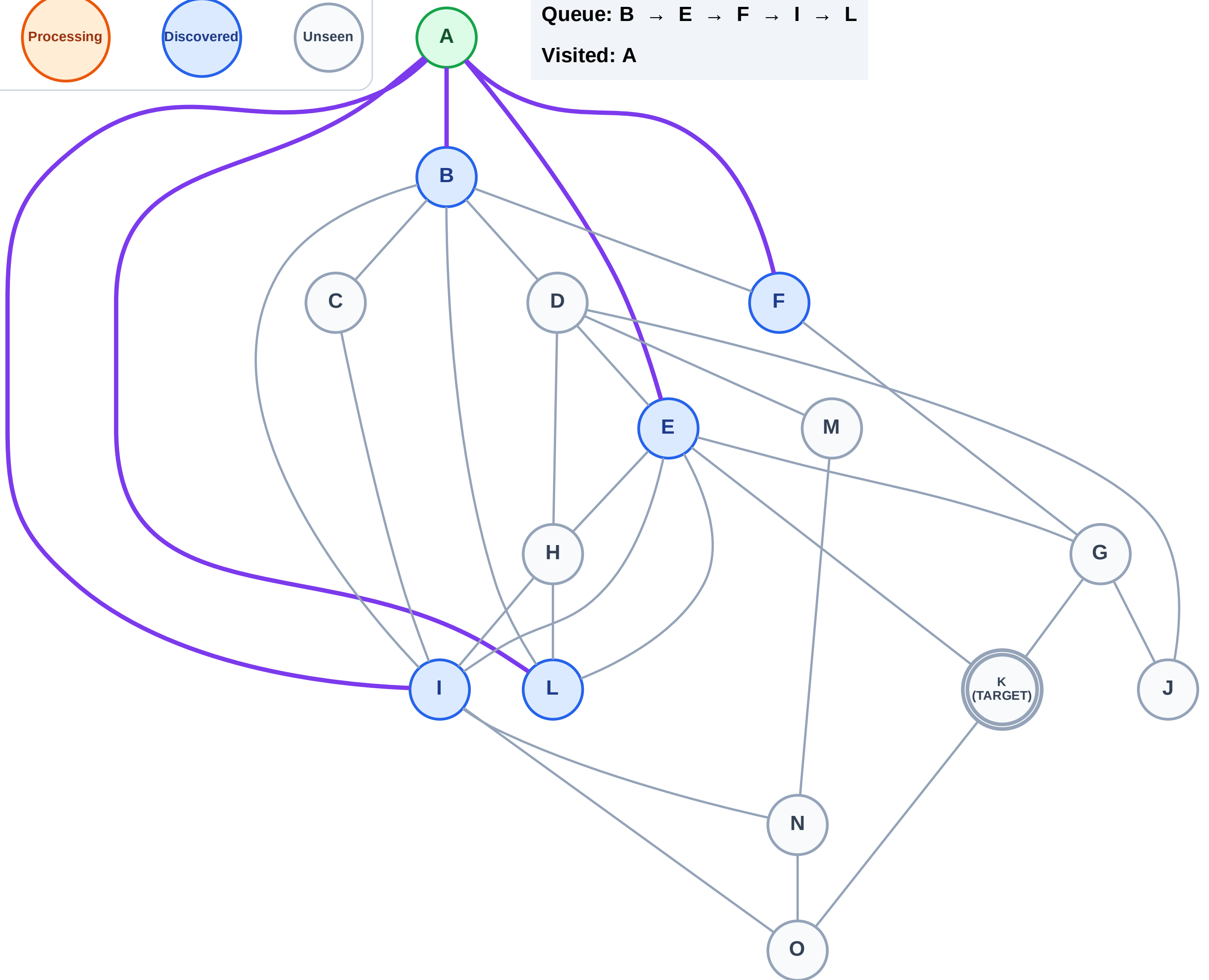
Discovered



Unseen

Queue: B → E → F → I → L

Visited: A



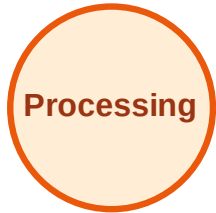
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



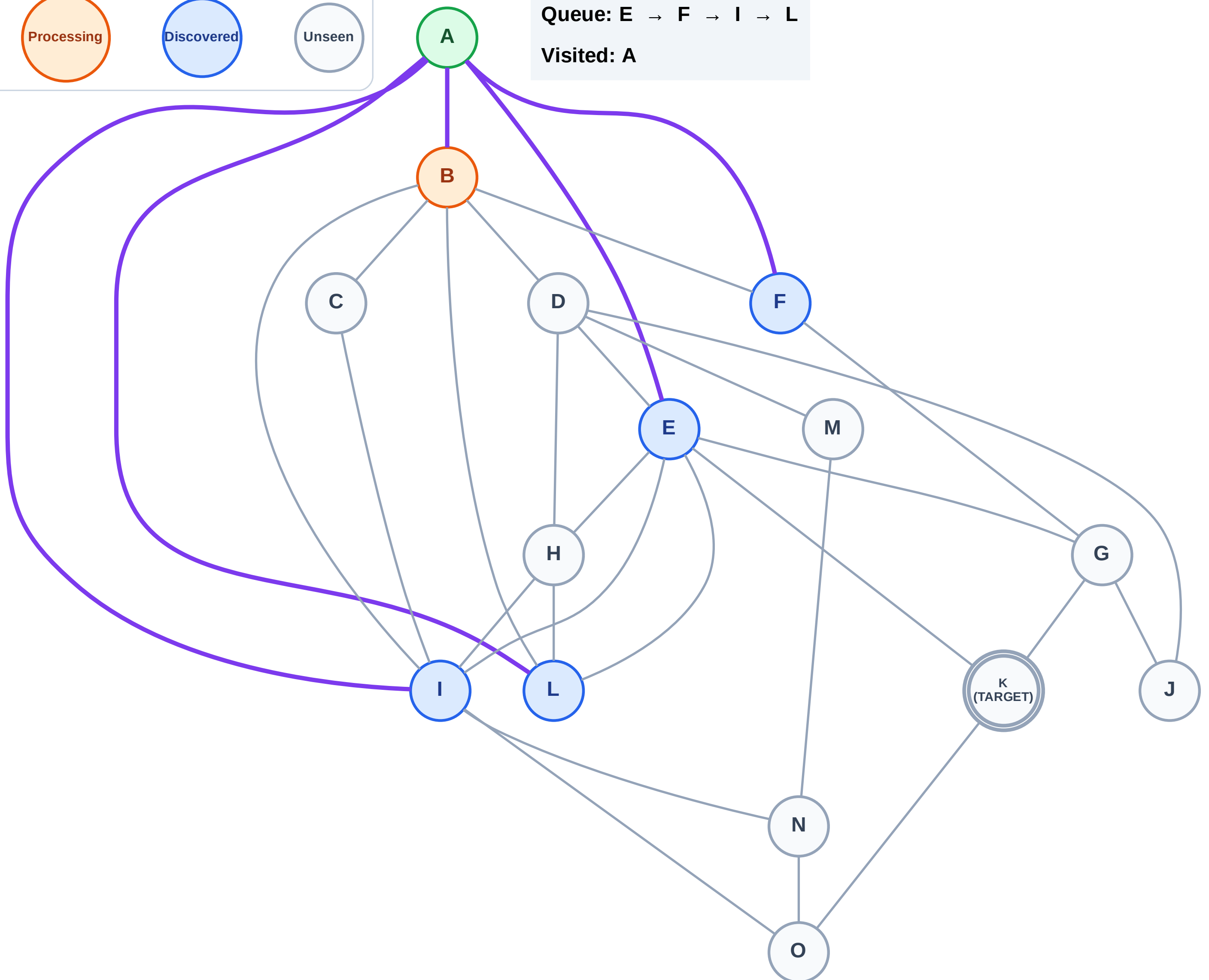
Discovered



Unseen

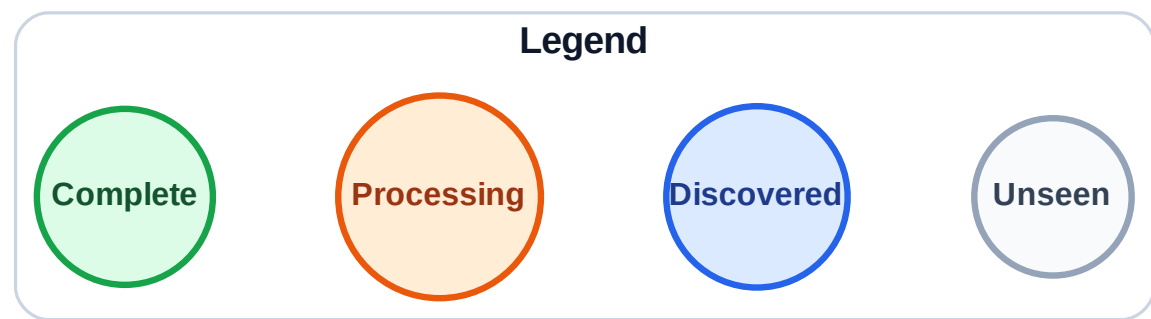
Queue: E → F → I → L

Visited: A



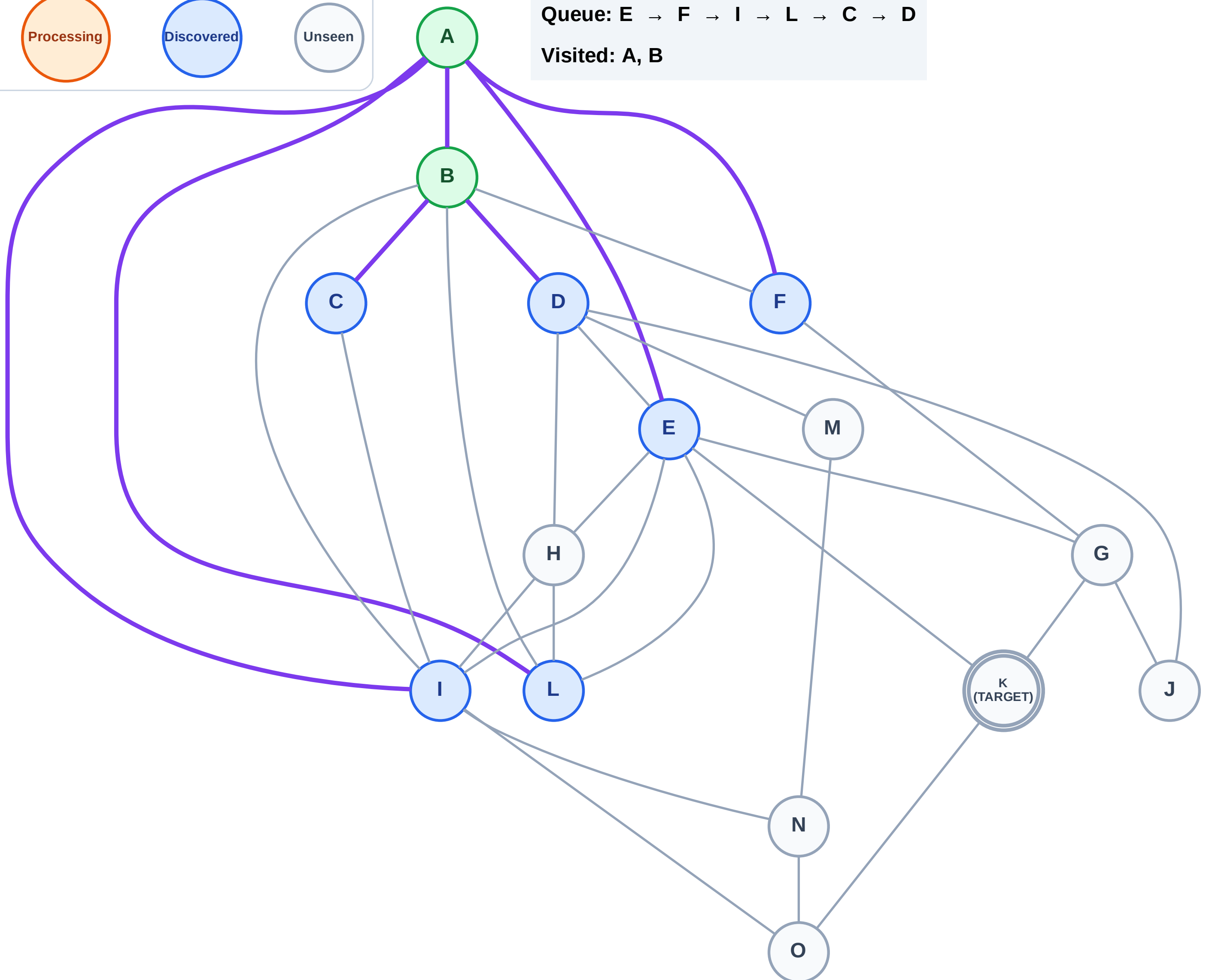
BFS Traversal

Step 5: Discover C, D from B



Queue: E → F → I → L → C → D

Visited: A, B



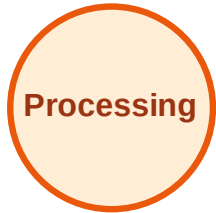
BFS Traversal

Step 6: Process node E

Legend



Complete



Processing



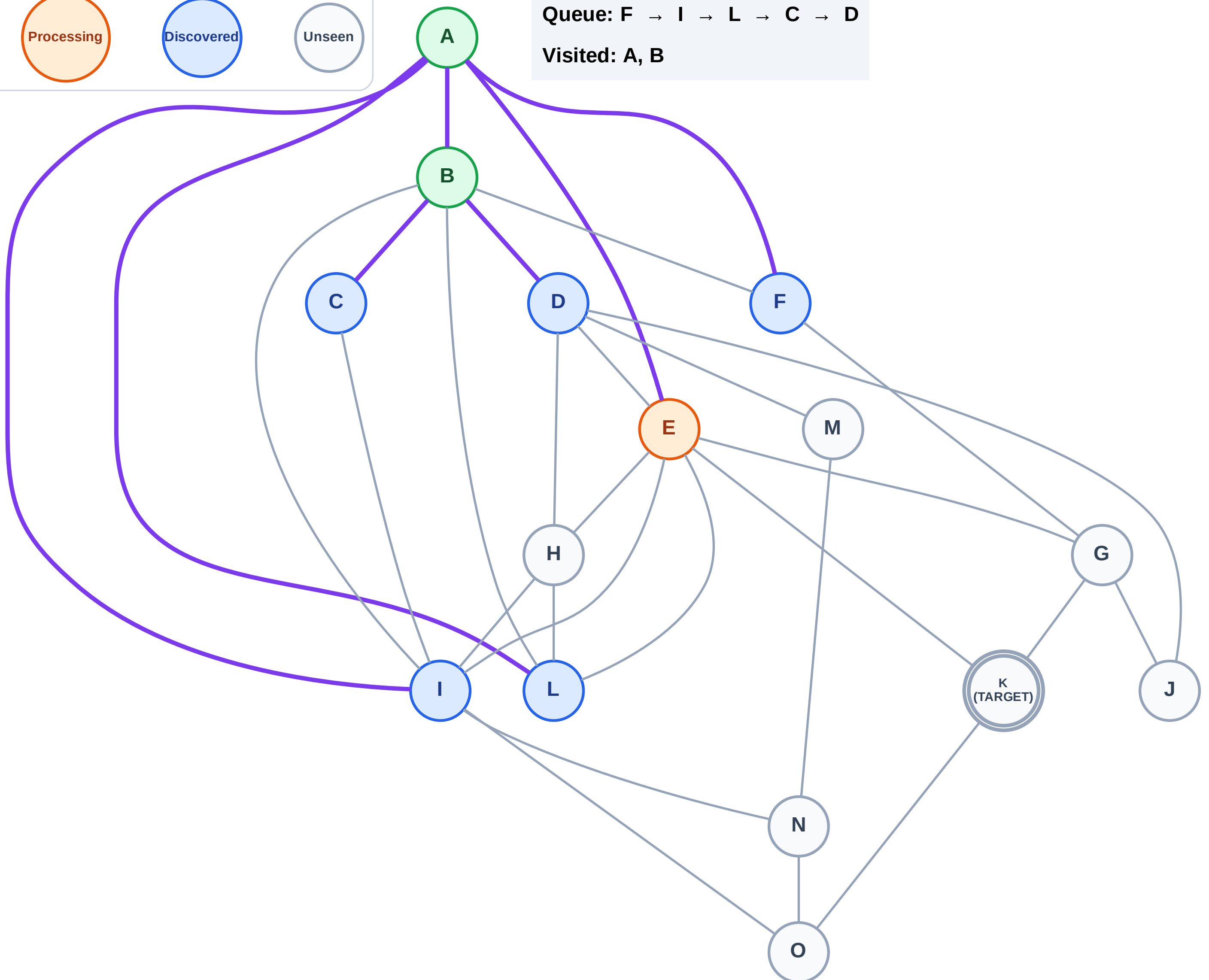
Discovered



Unseen

Queue: F → I → L → C → D

Visited: A, B

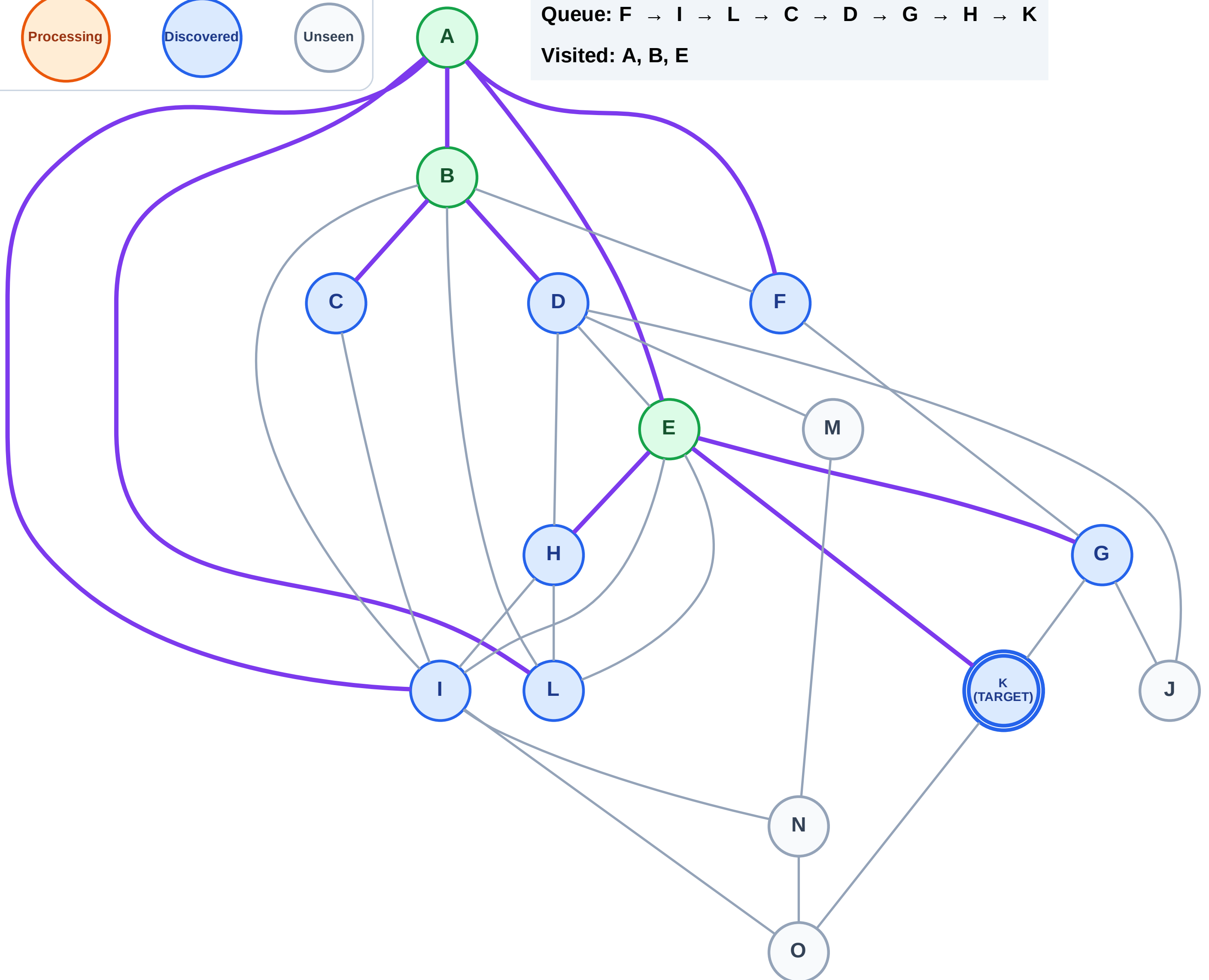


Step 7: Discover G, H, K from E

Queue: F → I → L → C → D → G → H → K

Legend

Unseen



BFS Traversal

Step 8: Process node F

Legend

Complete

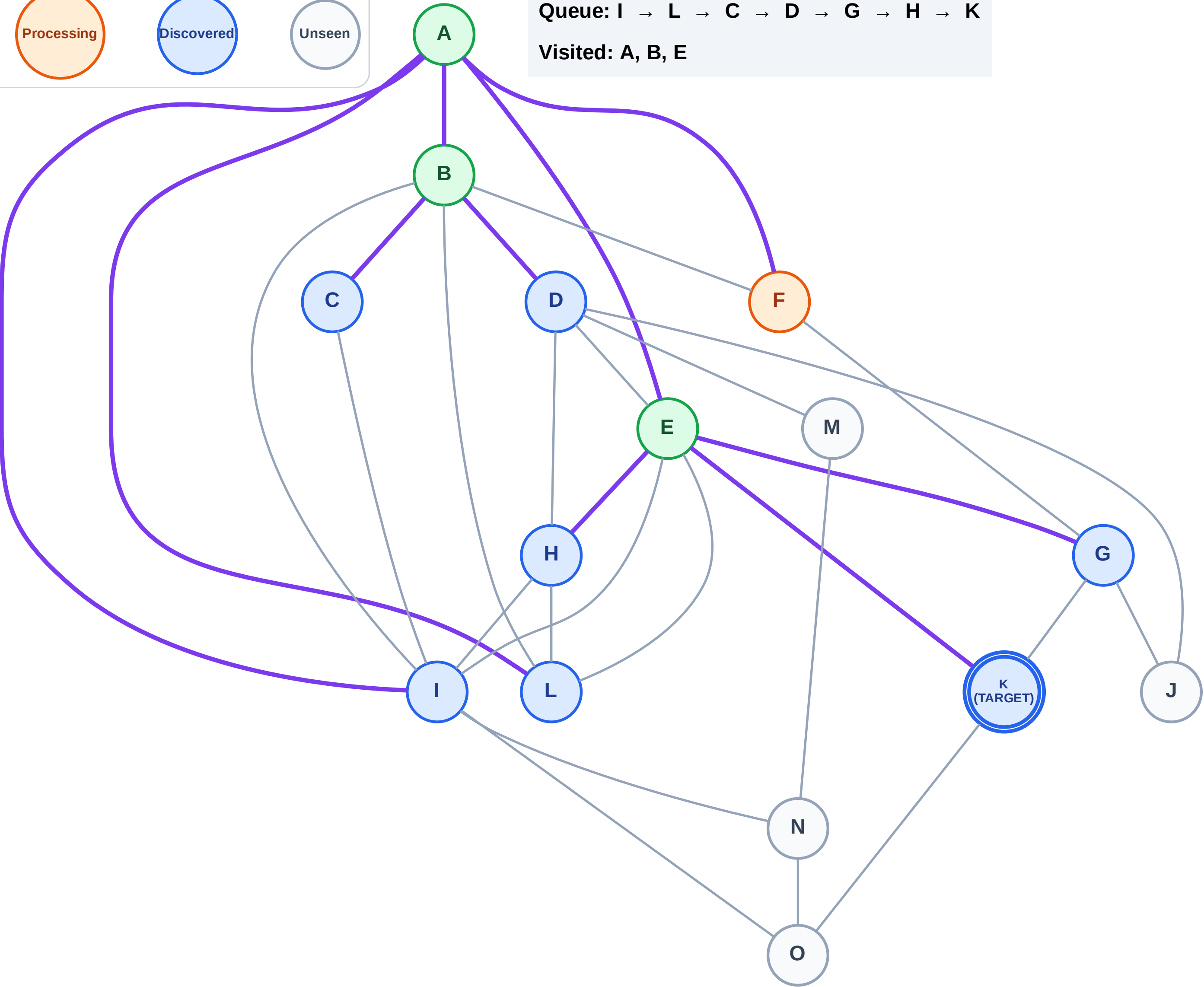
Processing

Discovered

Unseen

Queue: I → L → C → D → G → H → K

Visited: A, B, E



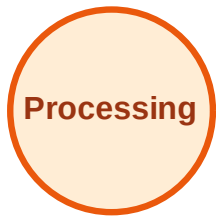
BFS Traversal

Step 9: No new neighbors from F

Legend



Complete



Processing



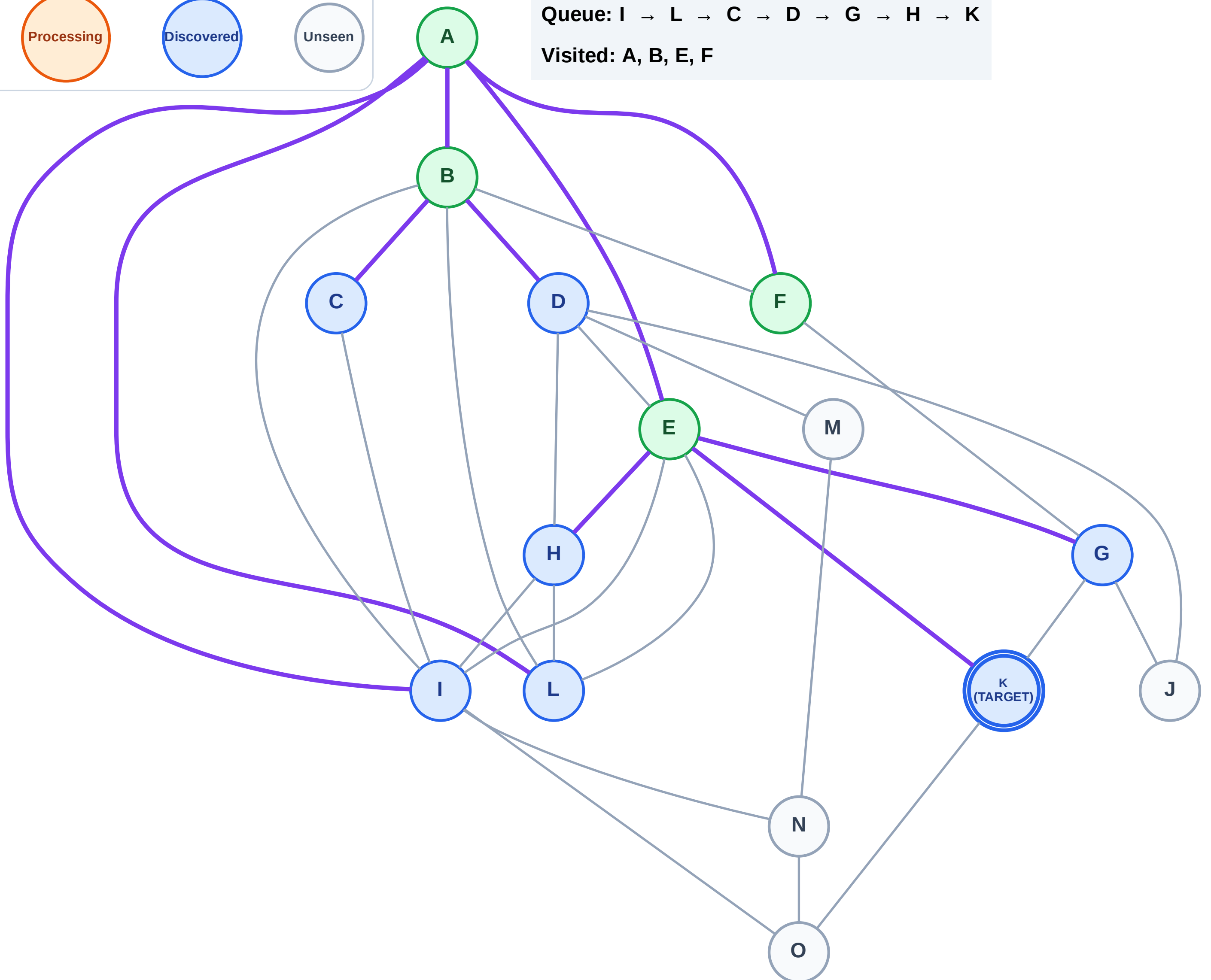
Discovered



Unseen

Queue: I → L → C → D → G → H → K

Visited: A, B, E, F



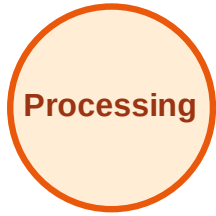
BFS Traversal

Step 10: Process node I

Legend



Complete



Processing



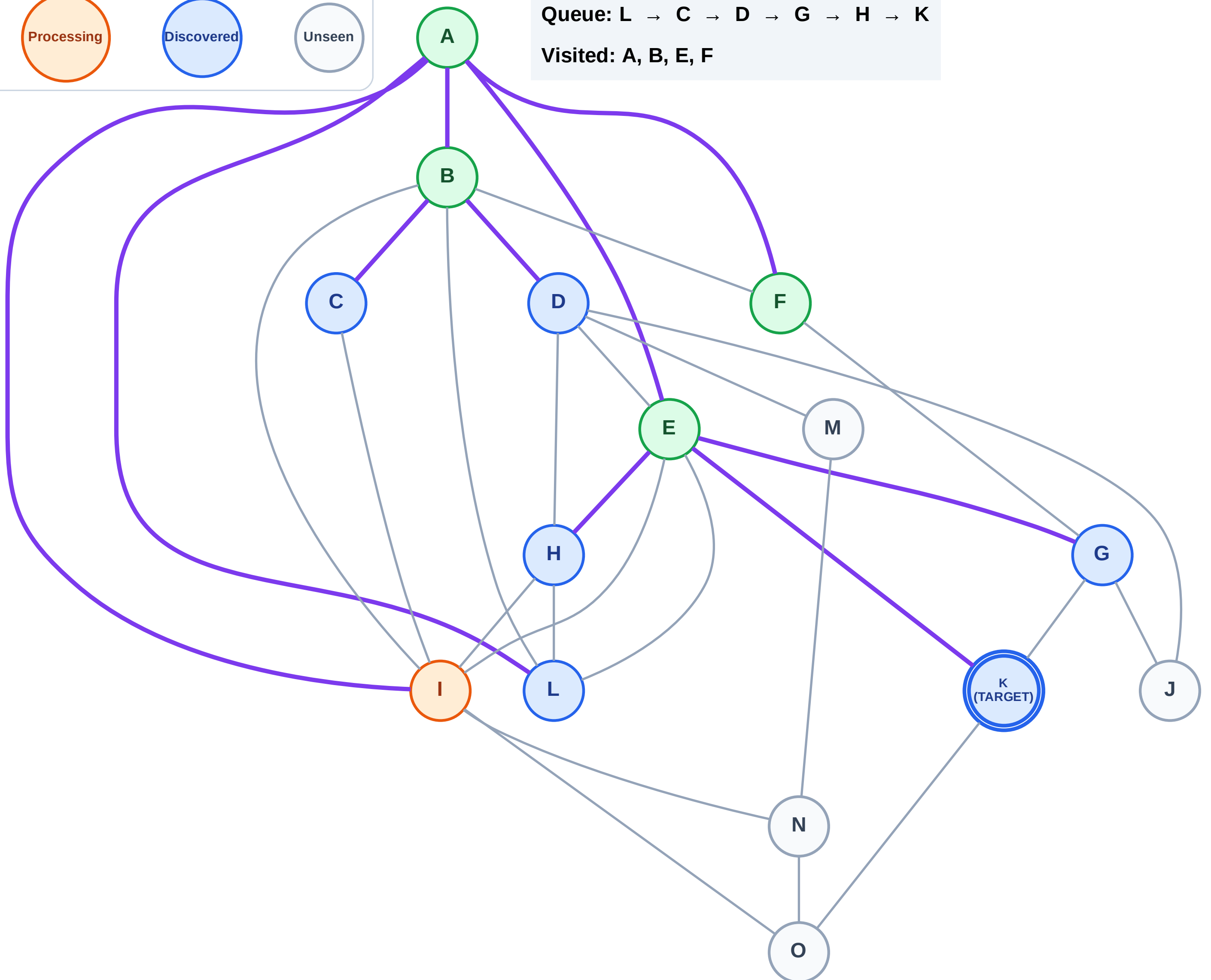
Discovered



Unseen

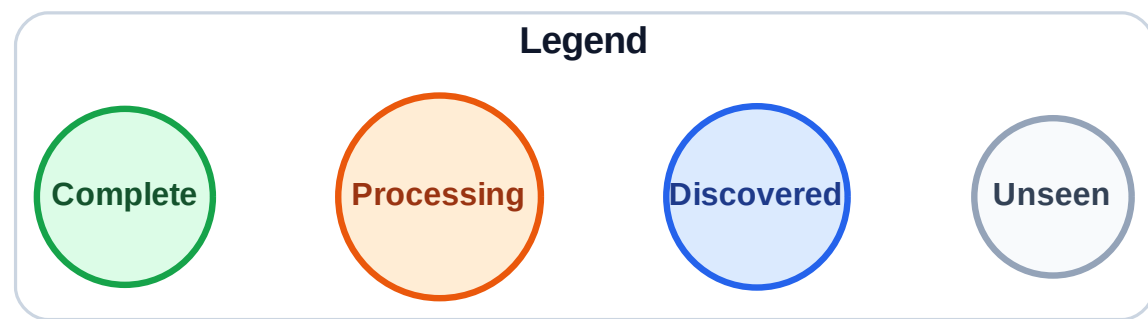
Queue: L → C → D → G → H → K

Visited: A, B, E, F



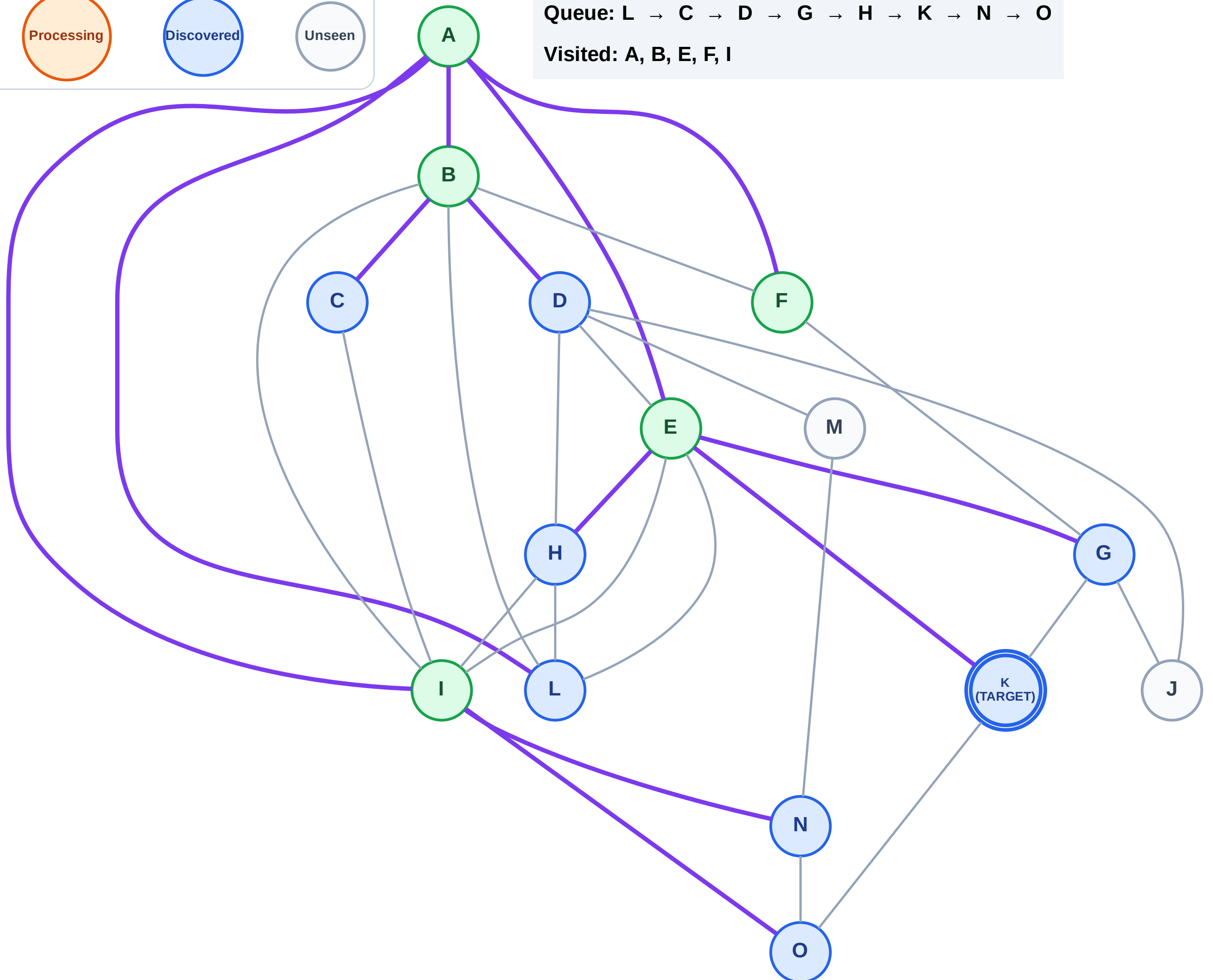
BFS Traversal

Step 11: Discover N, O from I



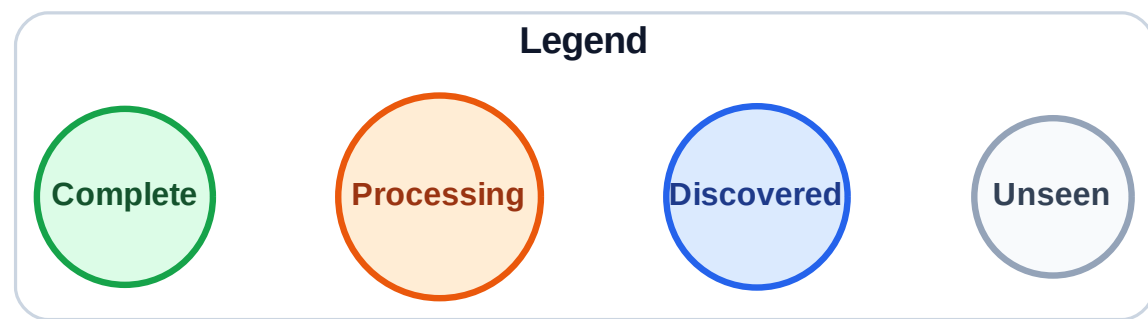
Queue: L → C → D → G → H → K → N → O

Visited: A, B, E, F, I



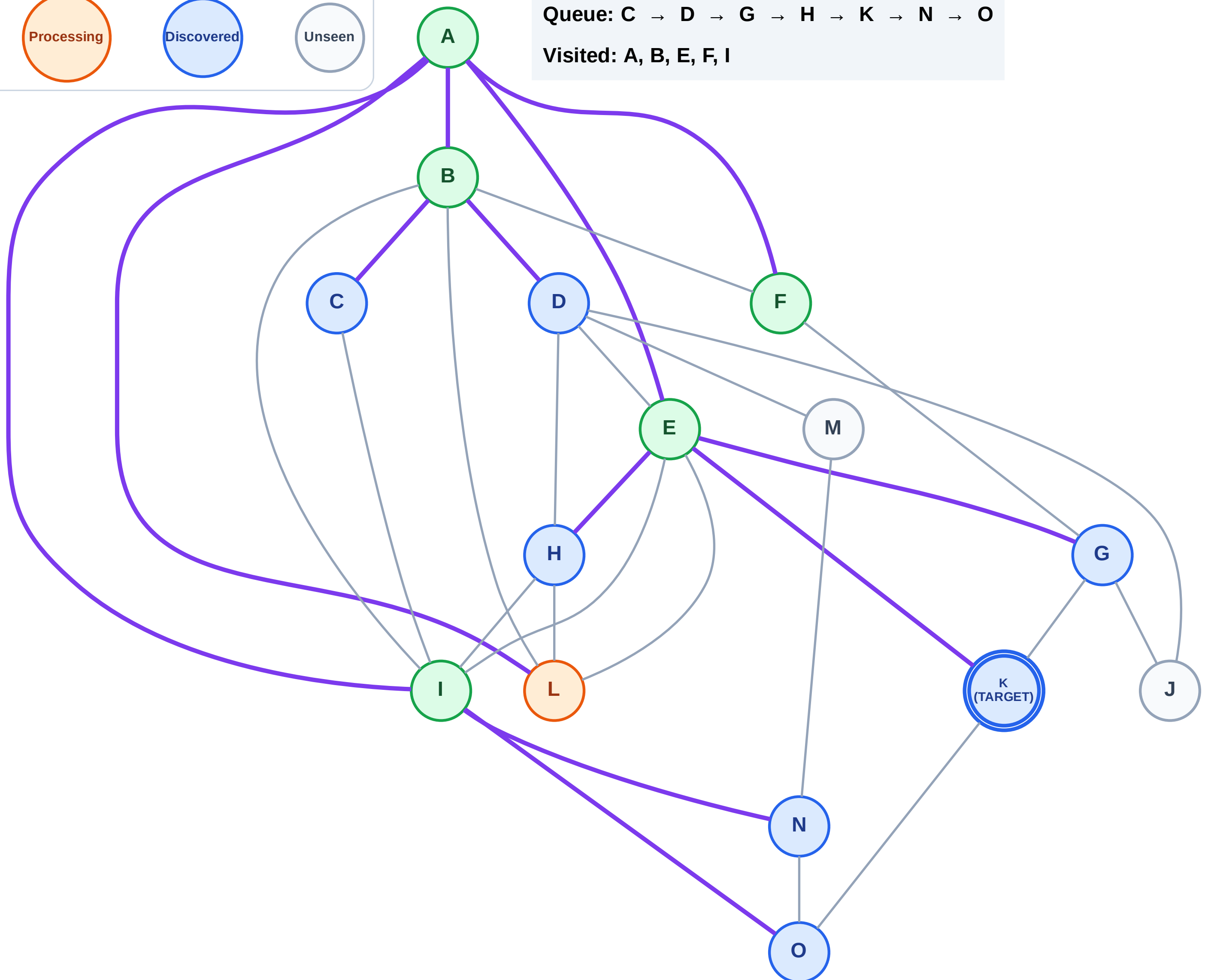
BFS Traversal

Step 12: Process node L



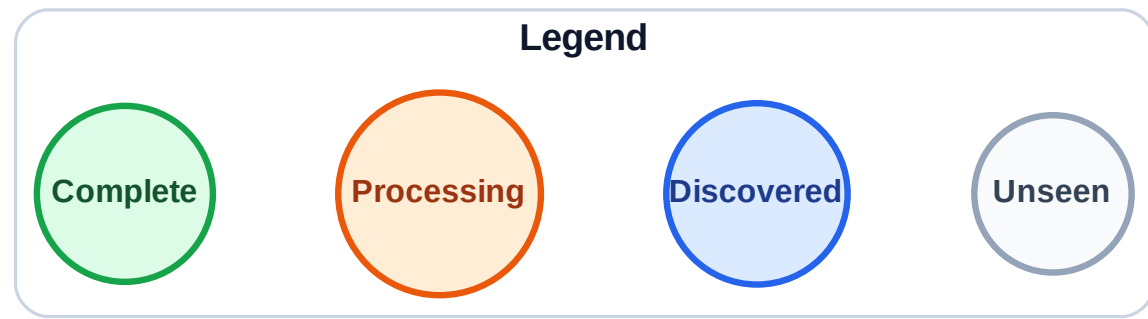
Queue: C → D → G → H → K → N → O

Visited: A, B, E, F, I



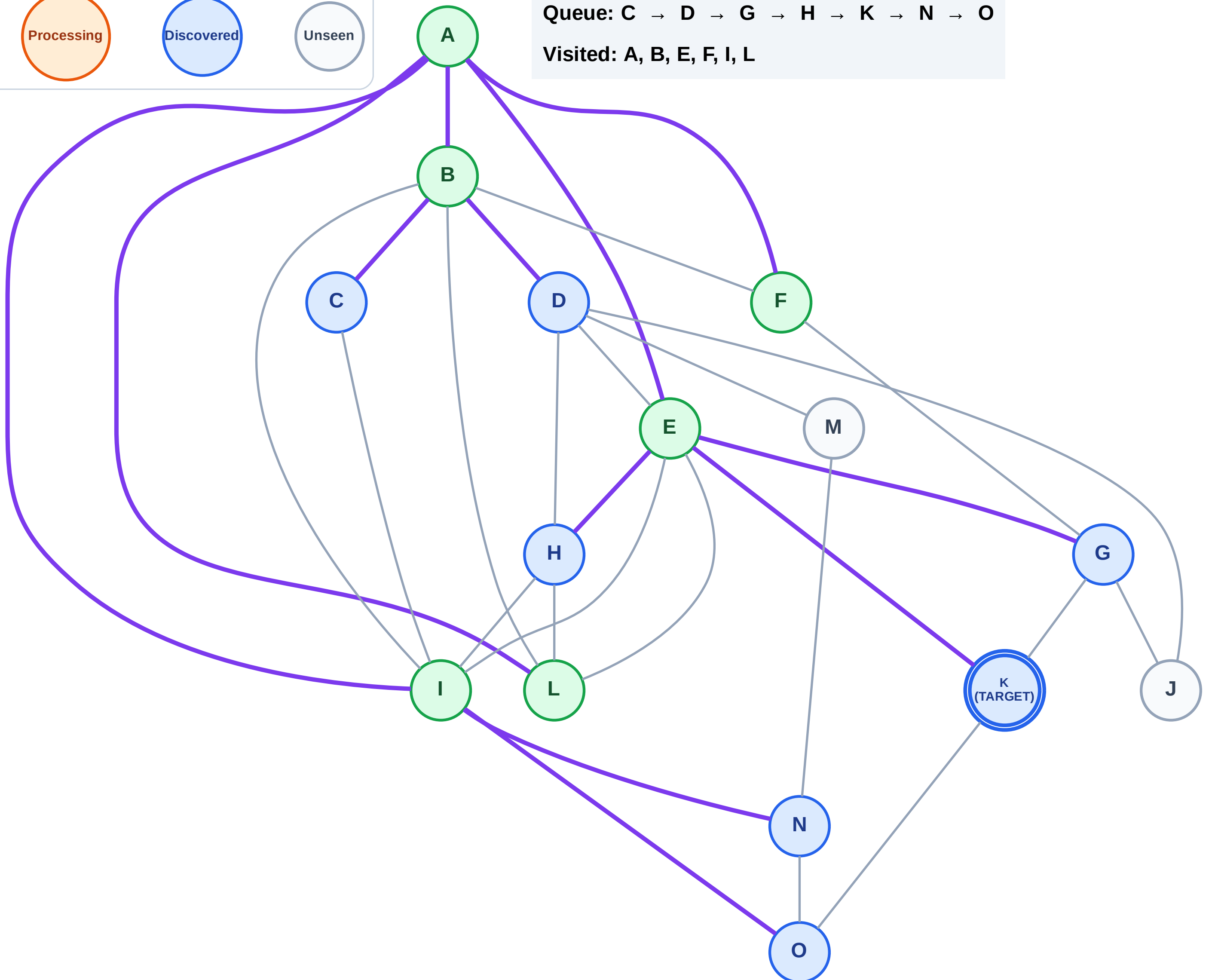
BFS Traversal

Step 13: No new neighbors from L



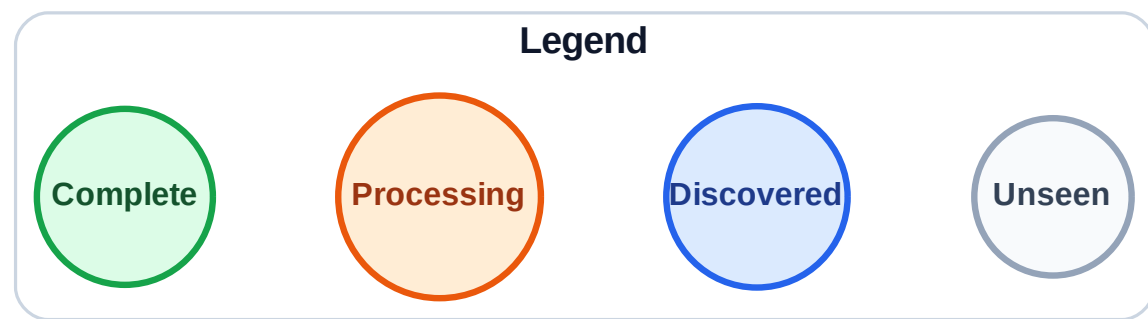
Queue: C → D → G → H → K → N → O

Visited: A, B, E, F, I, L



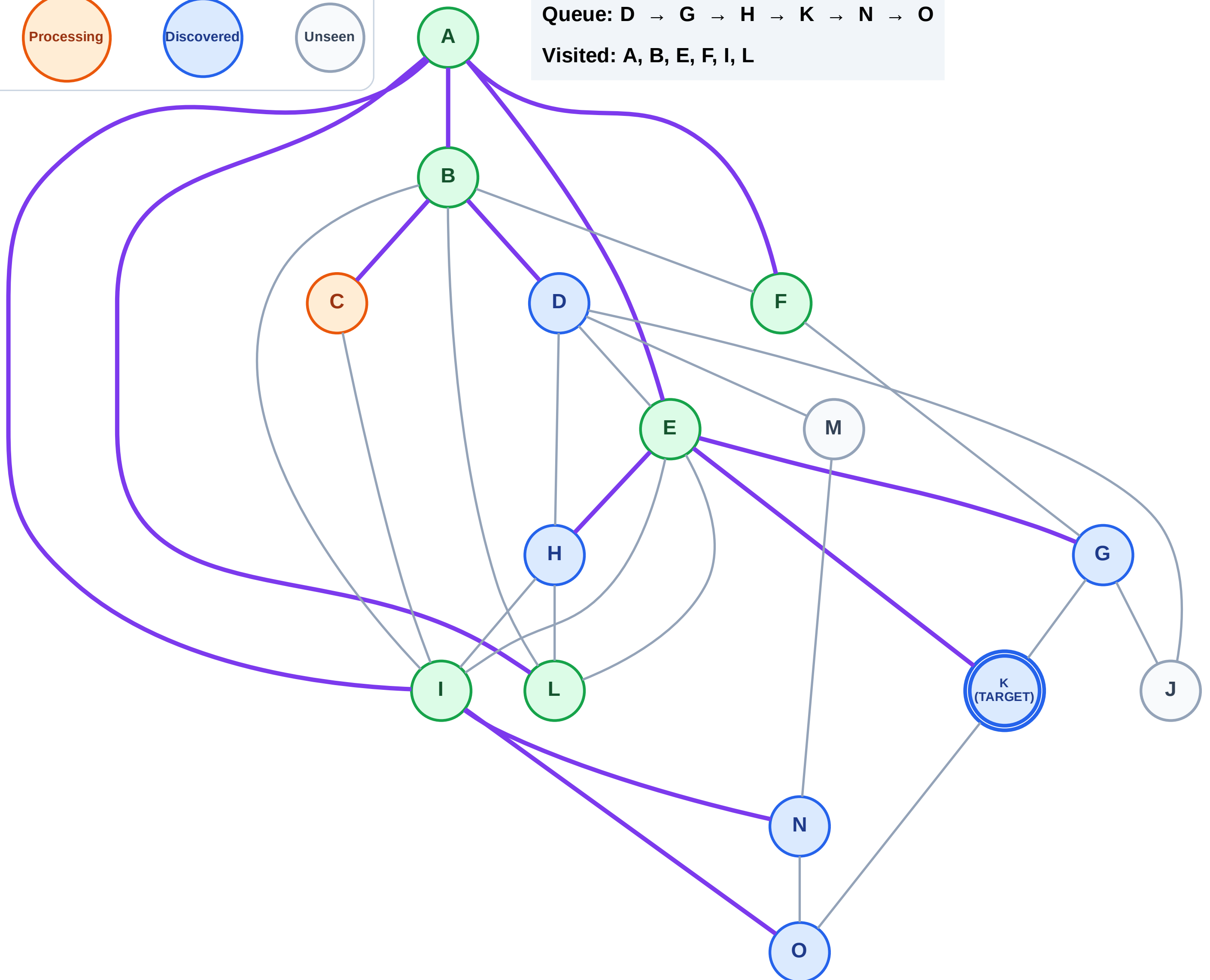
BFS Traversal

Step 14: Process node C



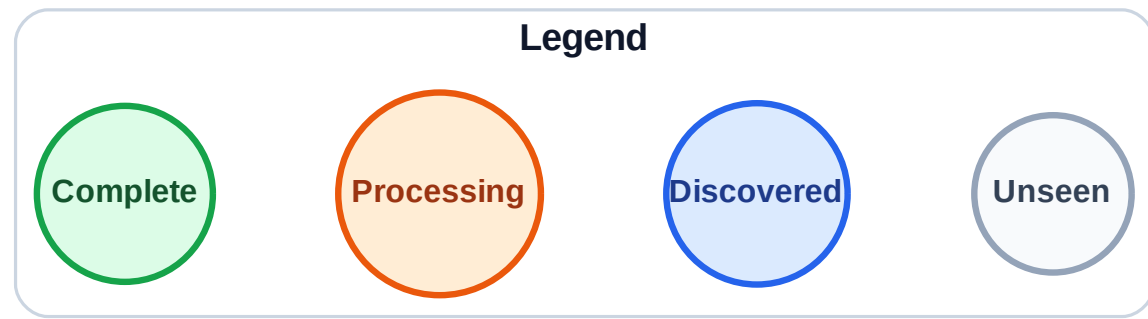
Queue: D → G → H → K → N → O

Visited: A, B, E, F, I, L



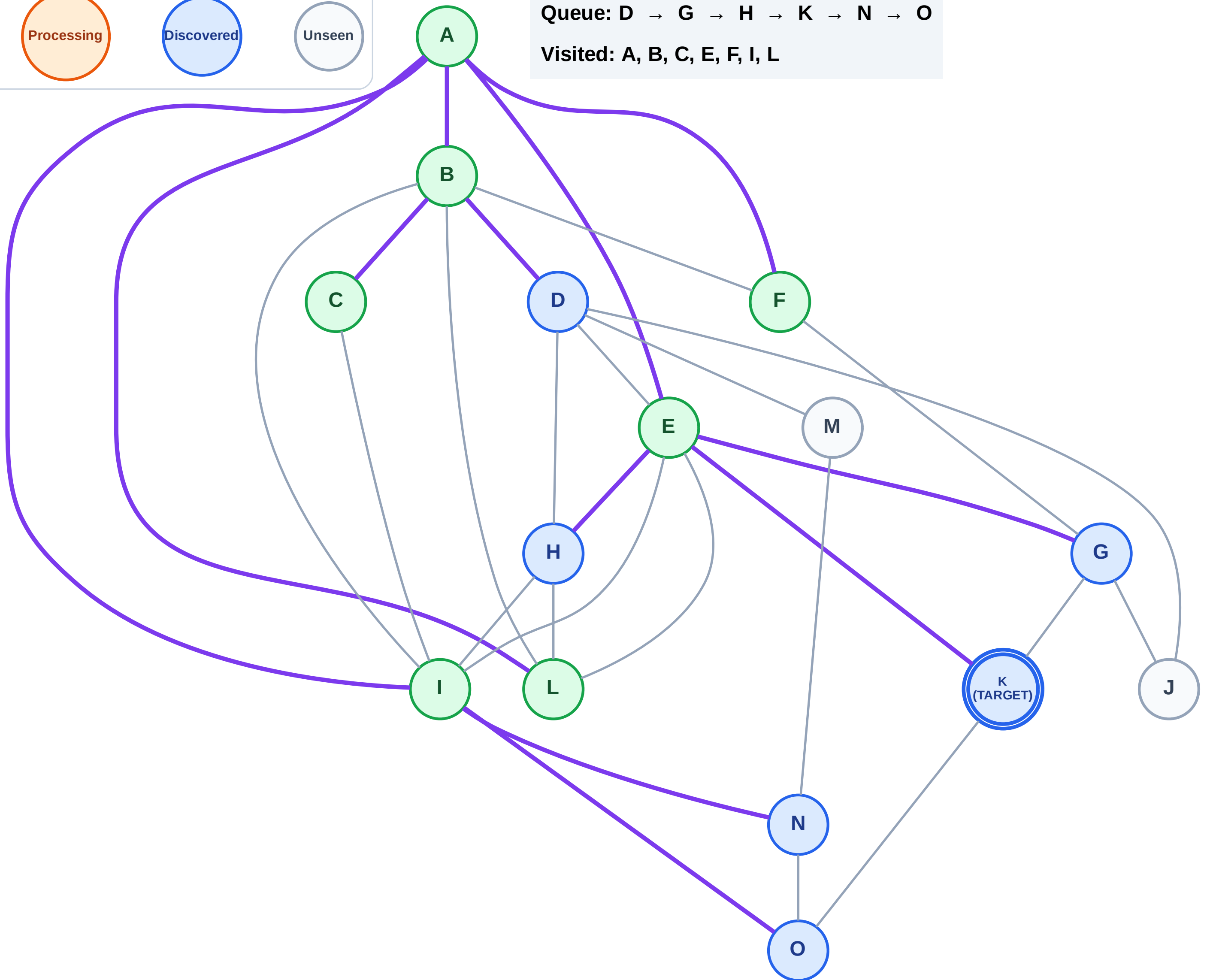
BFS Traversal

Step 15: No new neighbors from C



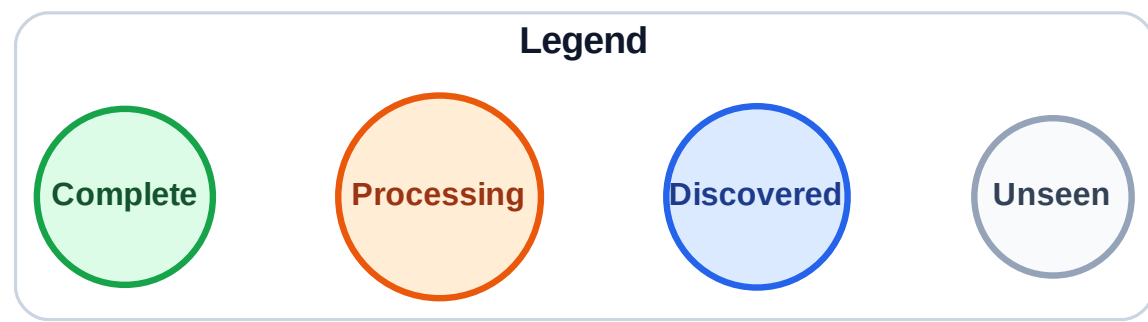
Queue: D → G → H → K → N → O

Visited: A, B, C, E, F, I, L



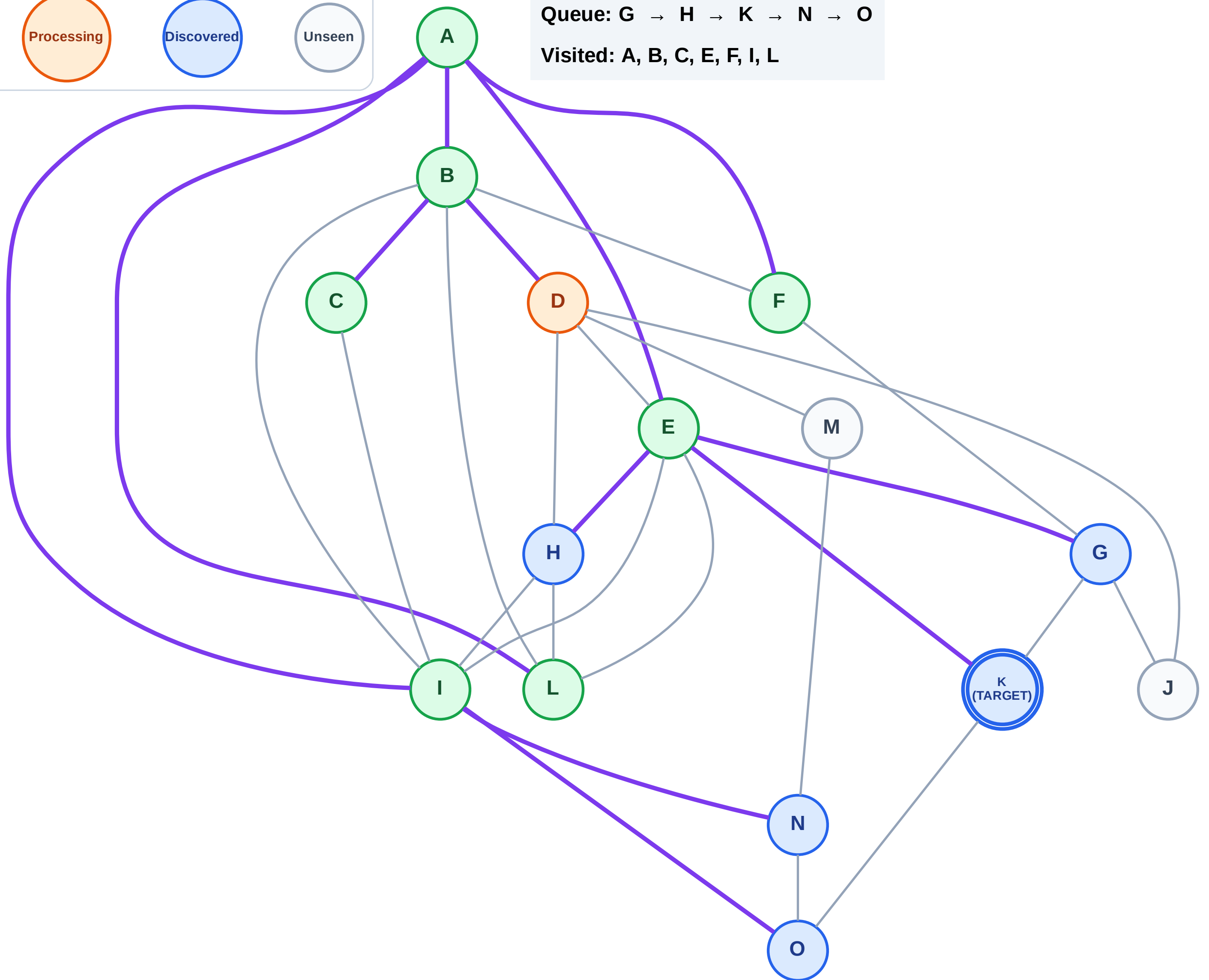
BFS Traversal

Step 16: Process node D



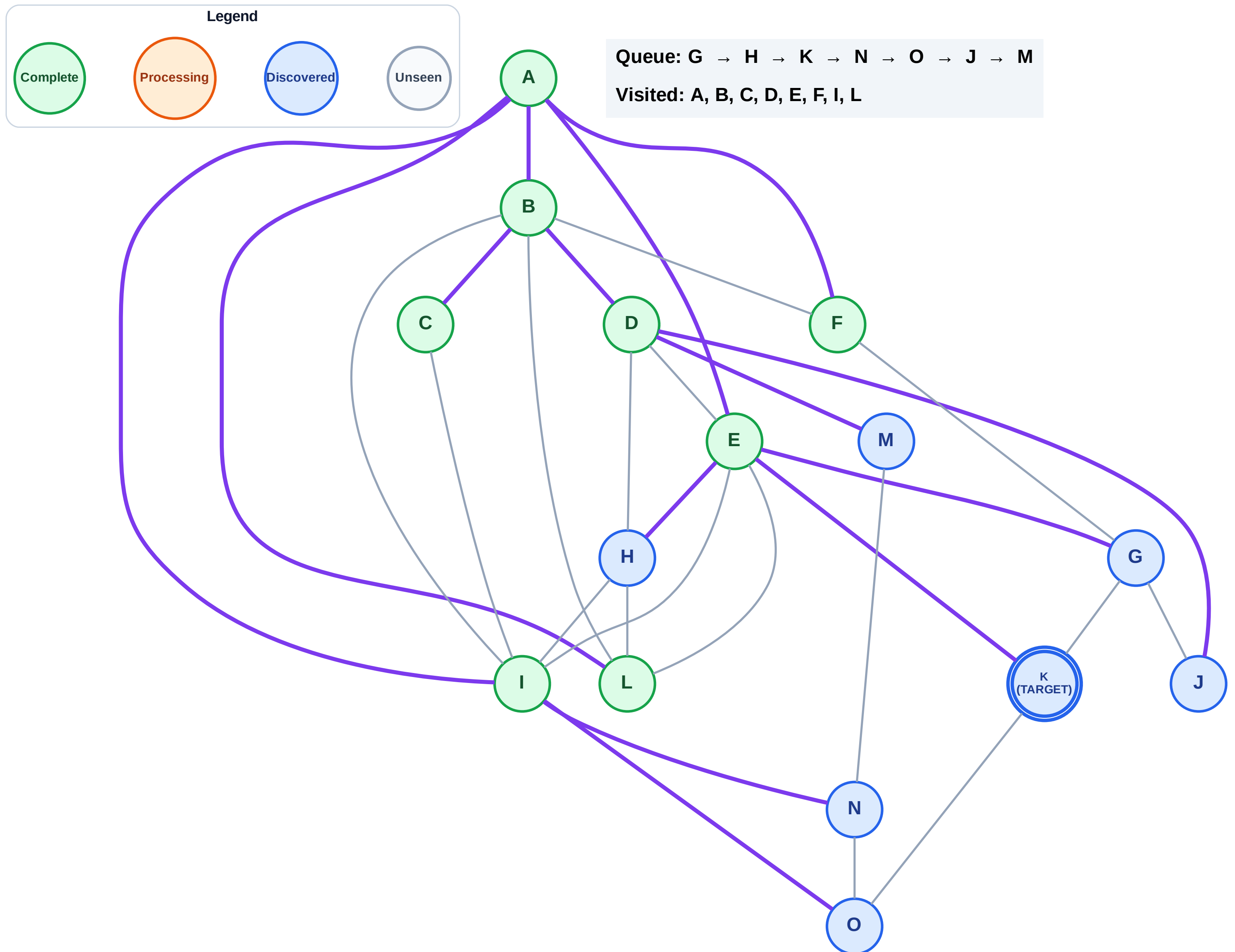
Queue: G → H → K → N → O

Visited: A, B, C, E, F, I, L



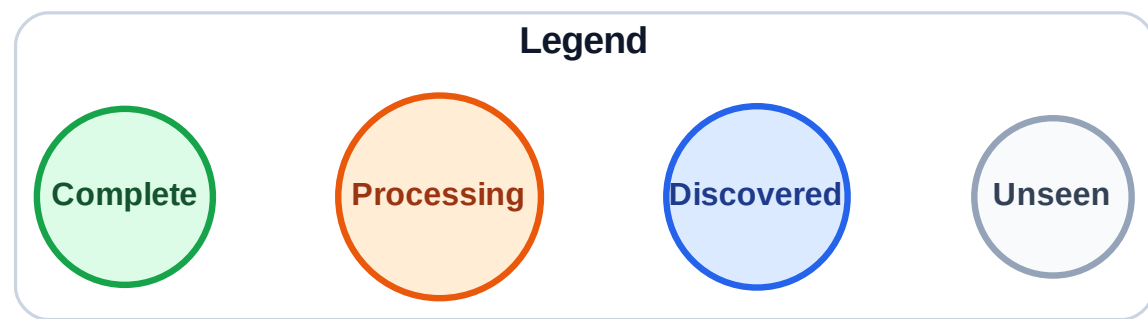
Step 17: Discover J, M from D

Step 17: Discover J, M from D



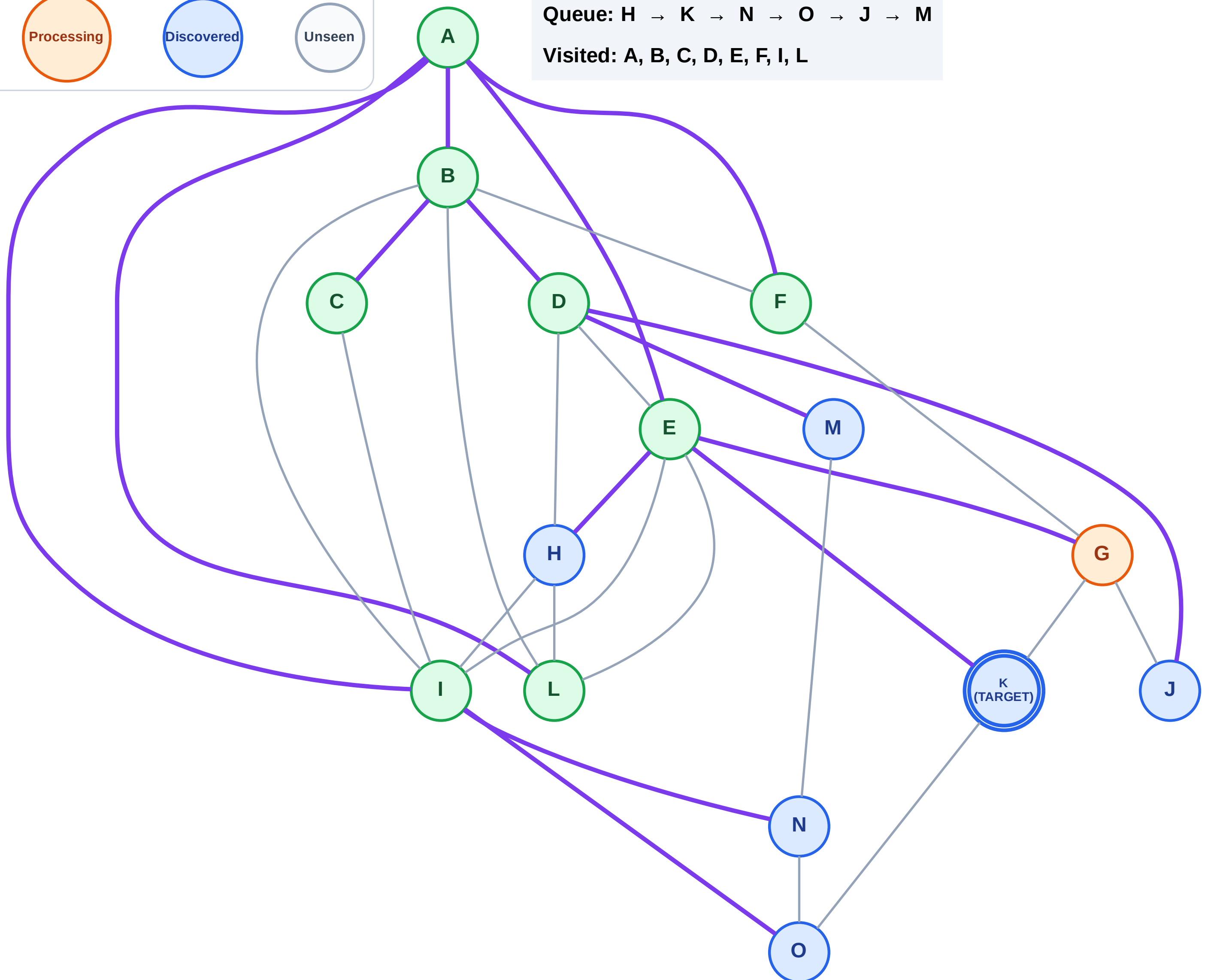
BFS Traversal

Step 18: Process node G



Queue: H → K → N → O → J → M

Visited: A, B, C, D, E, F, I, L



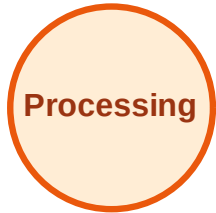
BFS Traversal

Step 19: No new neighbors from G

Legend



Complete



Processing



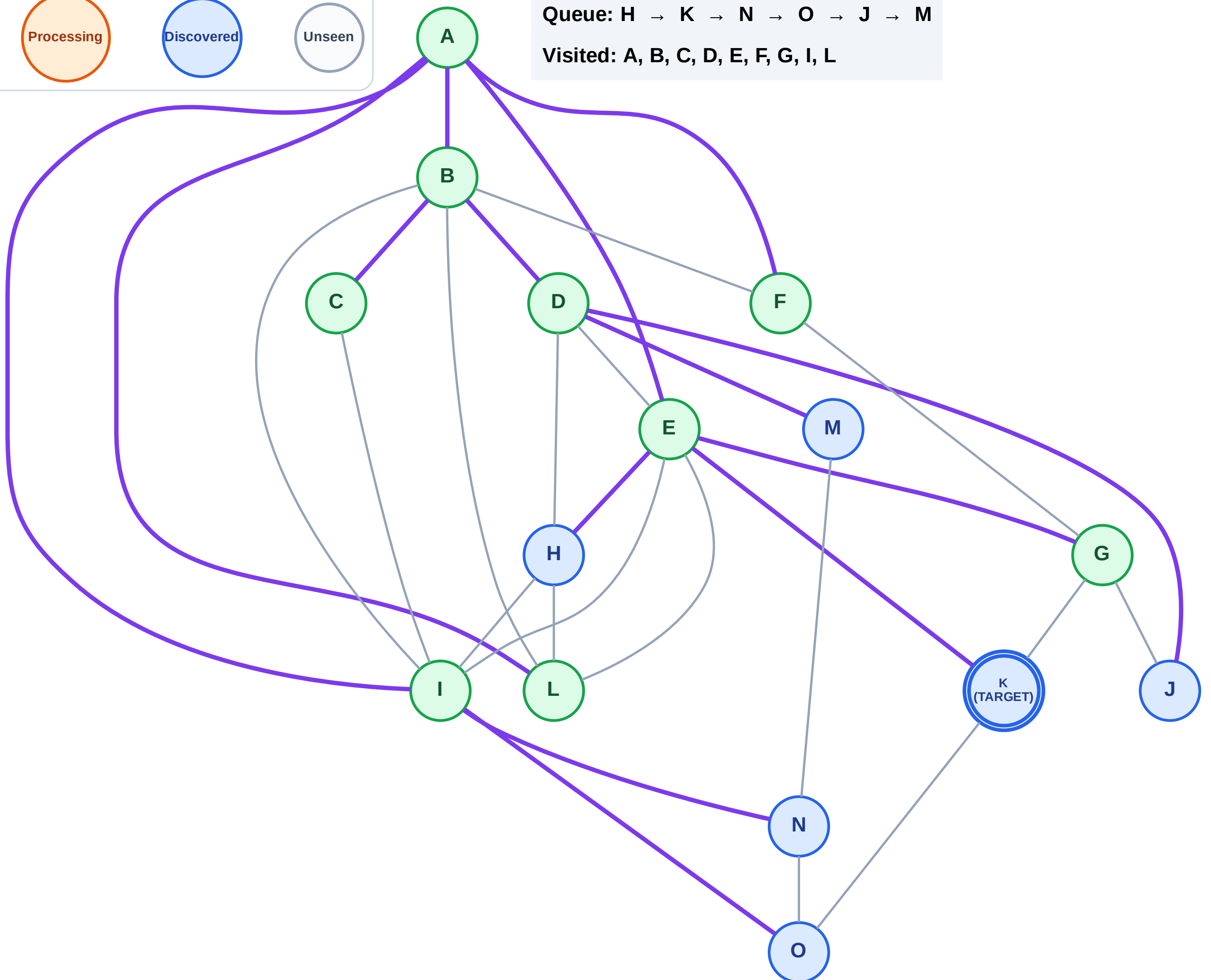
Discovered



Unseen

Queue: H → K → N → O → J → M

Visited: A, B, C, D, E, F, G, I, L



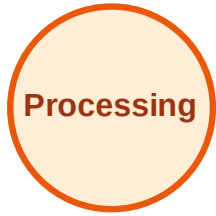
BFS Traversal

Step 20: Process node H

Legend



Complete



Processing



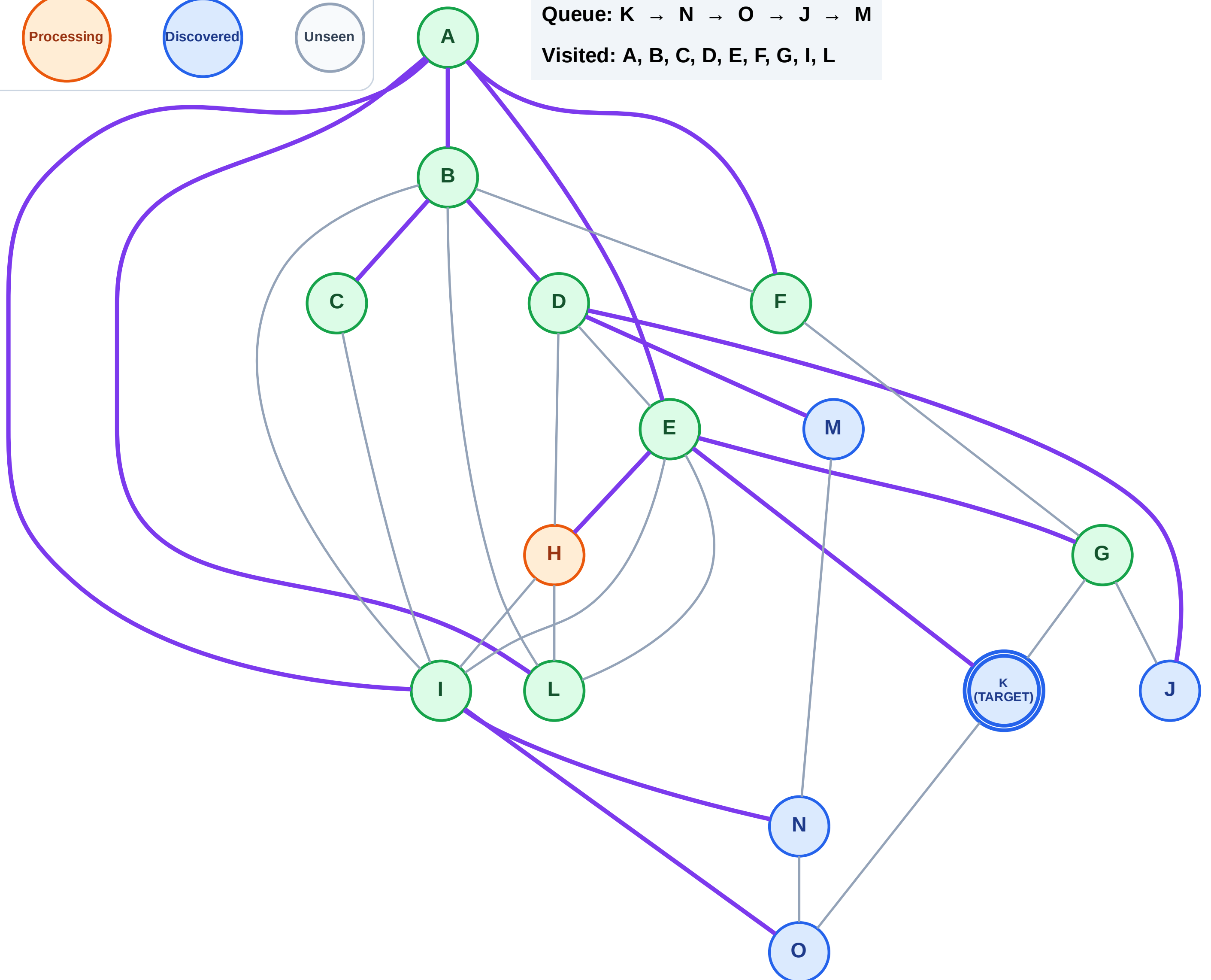
Discovered



Unseen

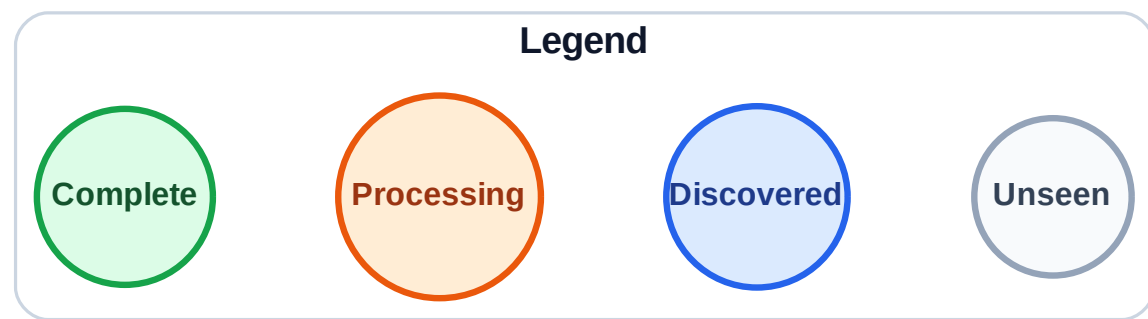
Queue: K → N → O → J → M

Visited: A, B, C, D, E, F, G, I, L



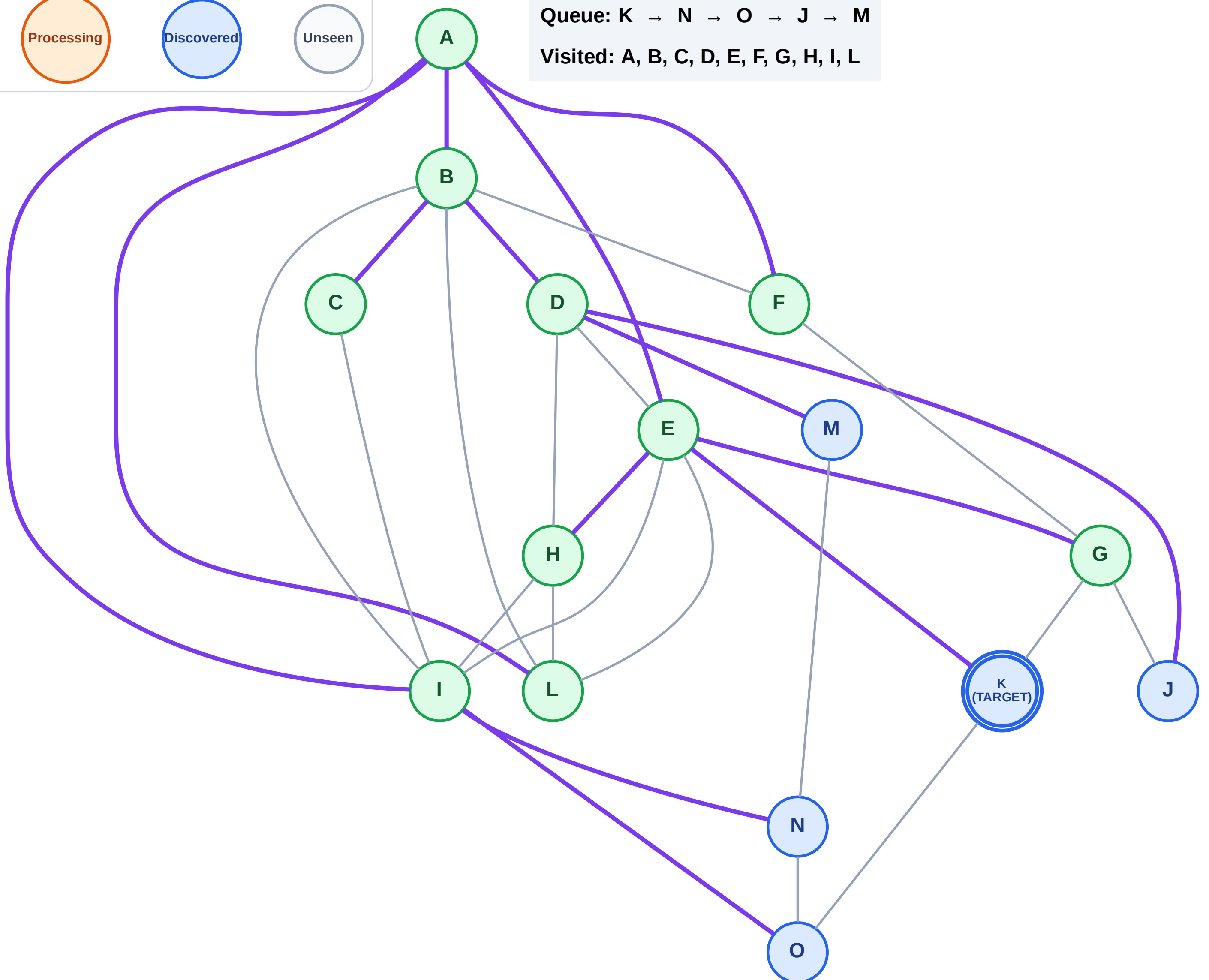
BFS Traversal

Step 21: No new neighbors from H



Queue: K → N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, L



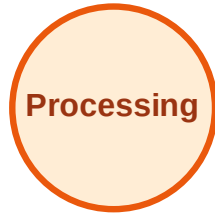
BFS Traversal

Step 22: Process node K

Legend



Complete



Processing



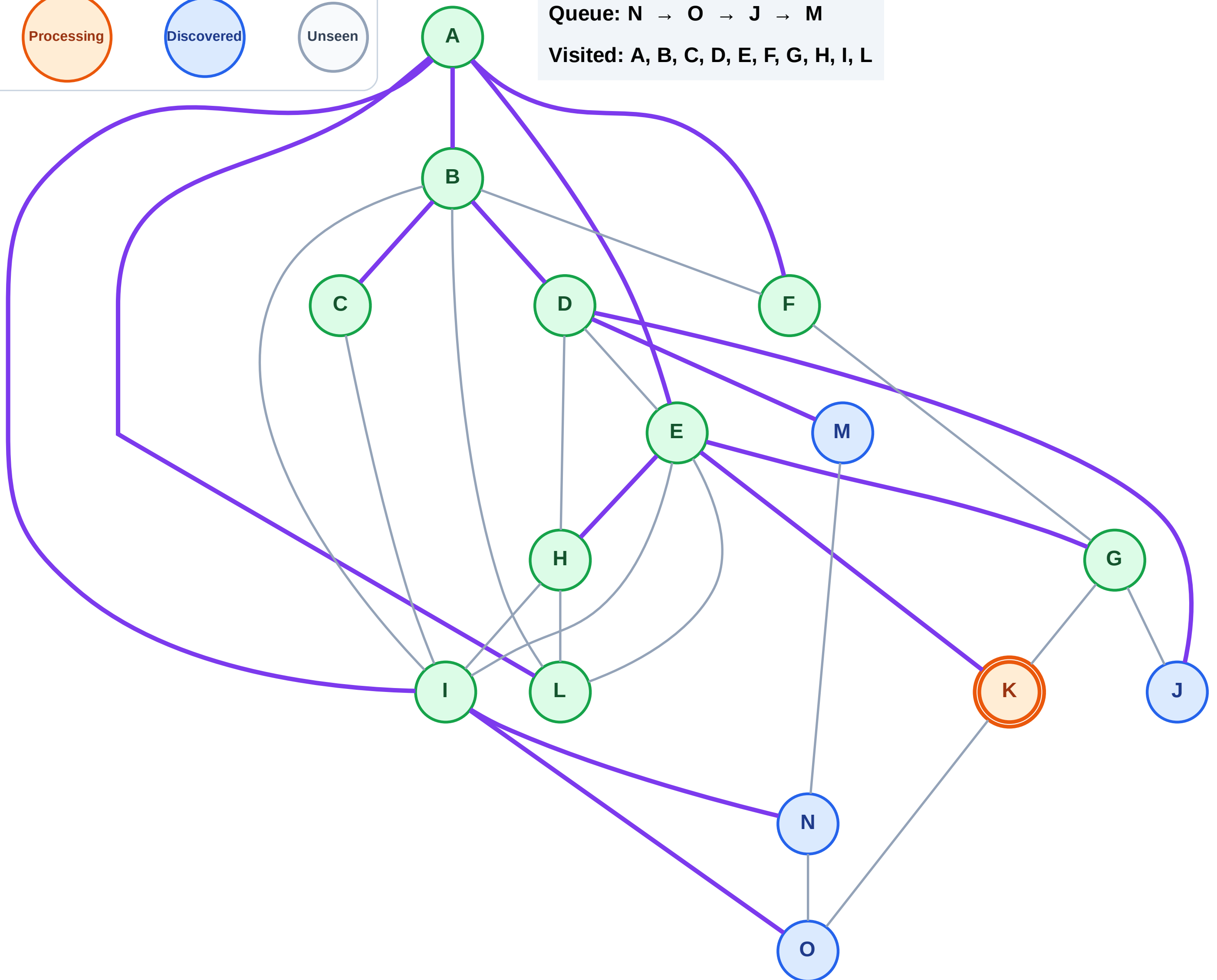
Discovered



Unseen

Queue: N → O → J → M

Visited: A, B, C, D, E, F, G, H, I, L



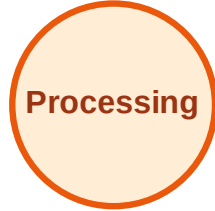
BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend



Complete



Processing



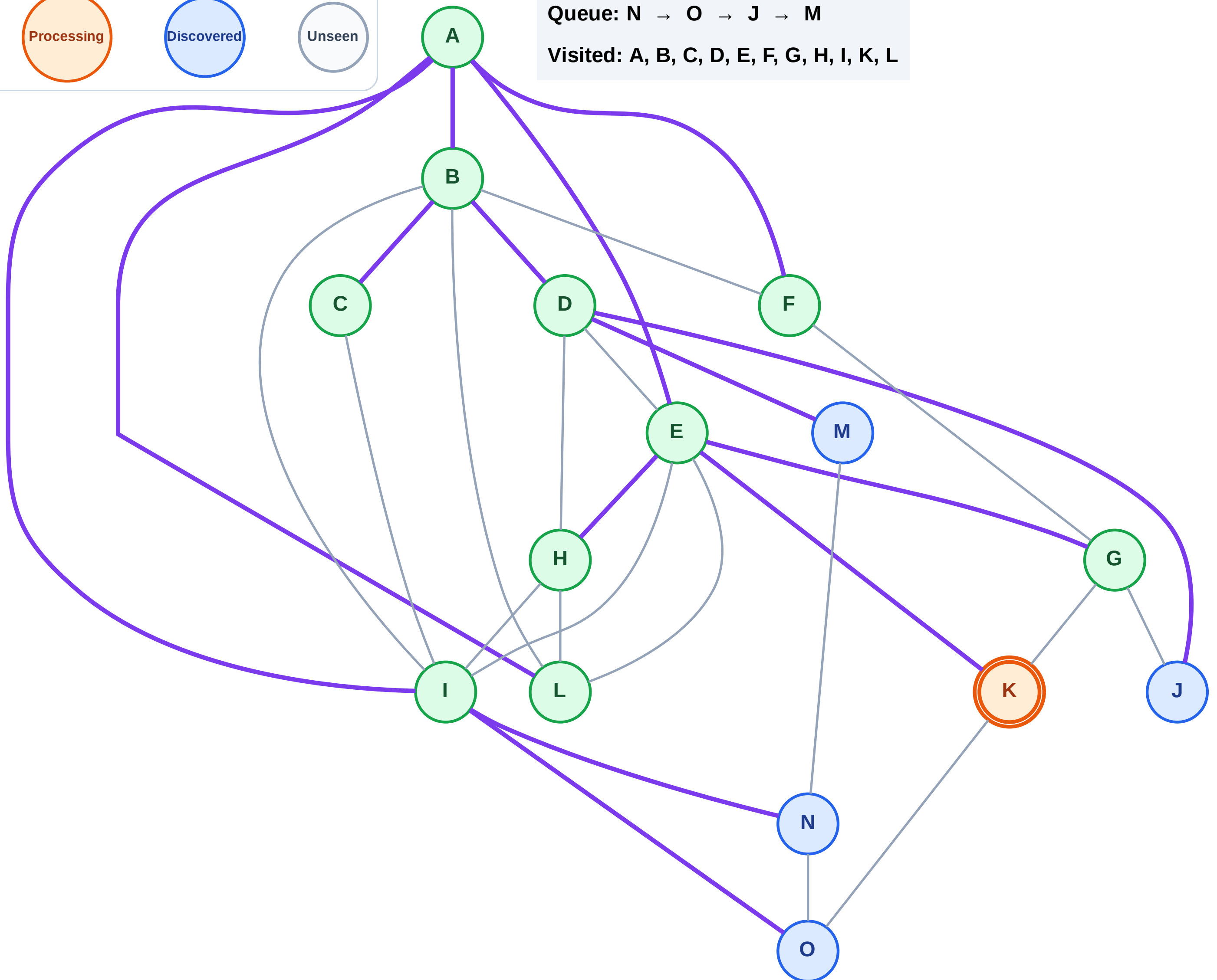
Discovered



Unseen

Queue: N → O → J → M

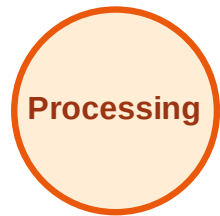
Visited: A, B, C, D, E, F, G, H, I, K, L



DFS Traversal

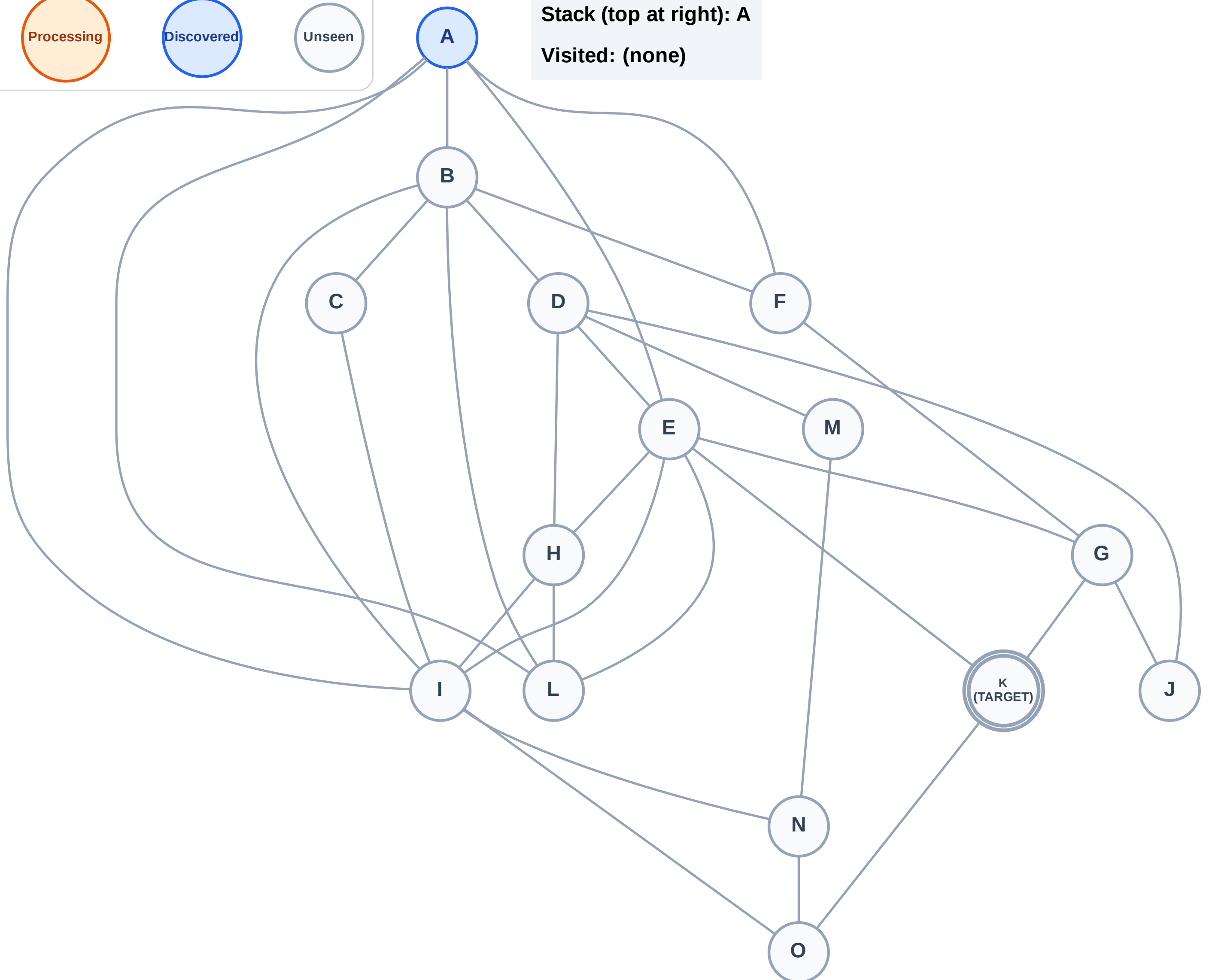
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

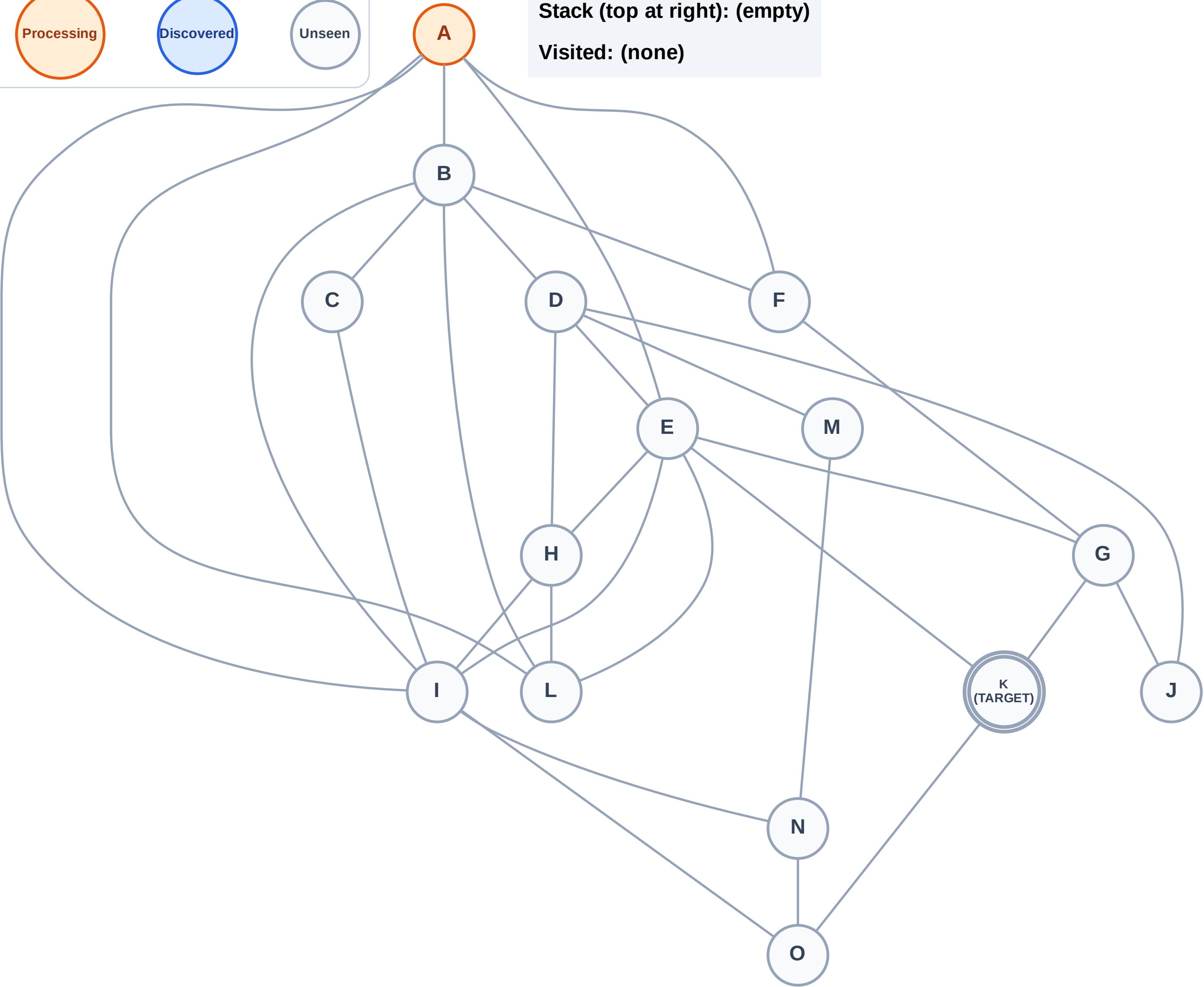
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



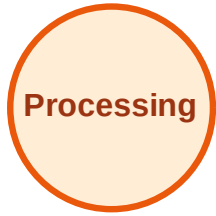
DFS Traversal

Step 3: Discover L, I, F, E, B from A

Legend



Complete



Processing



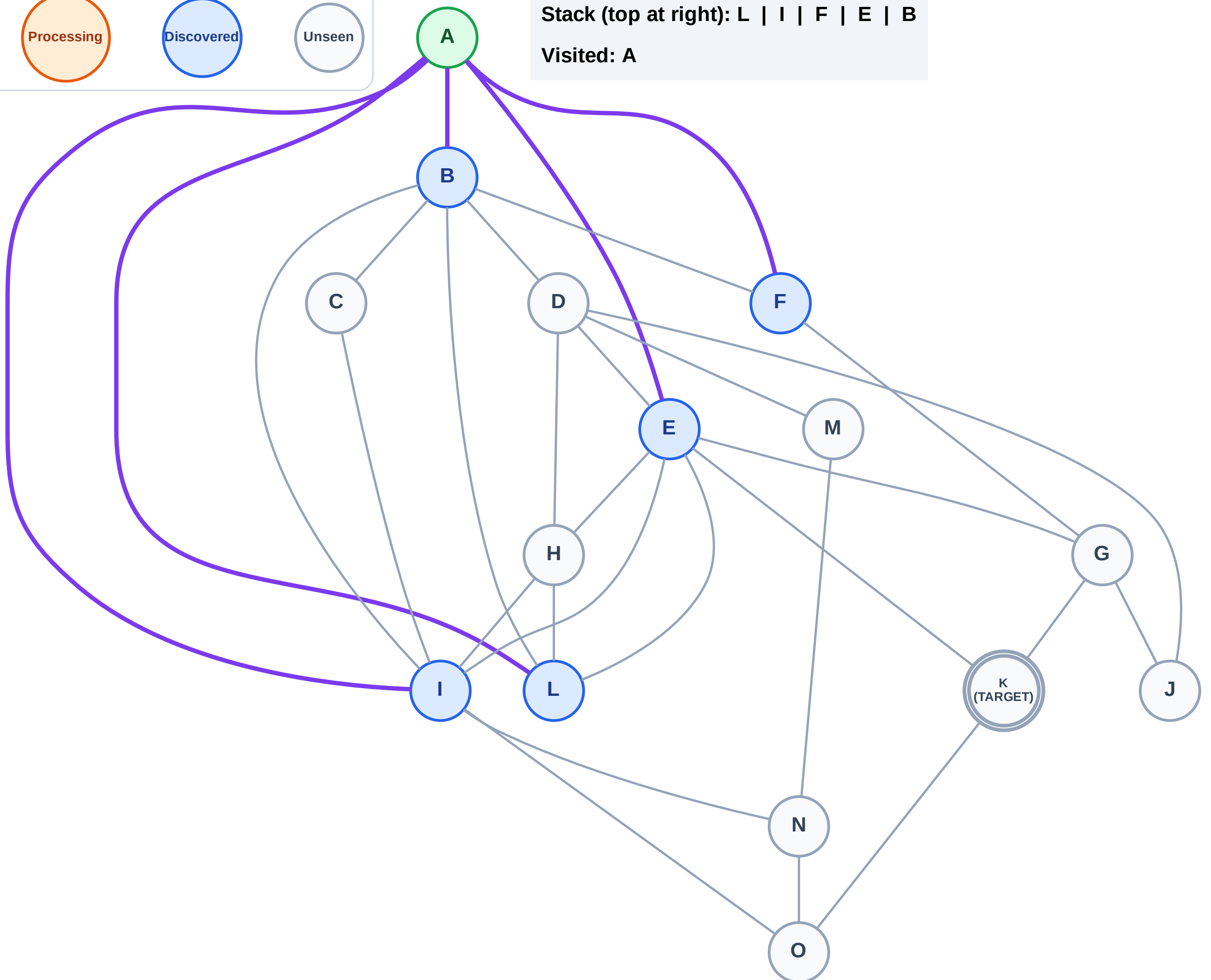
Discovered



Unseen

Stack (top at right): L | I | F | E | B

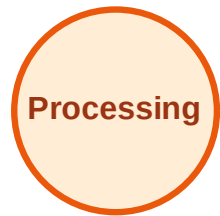
Visited: A



DFS Traversal

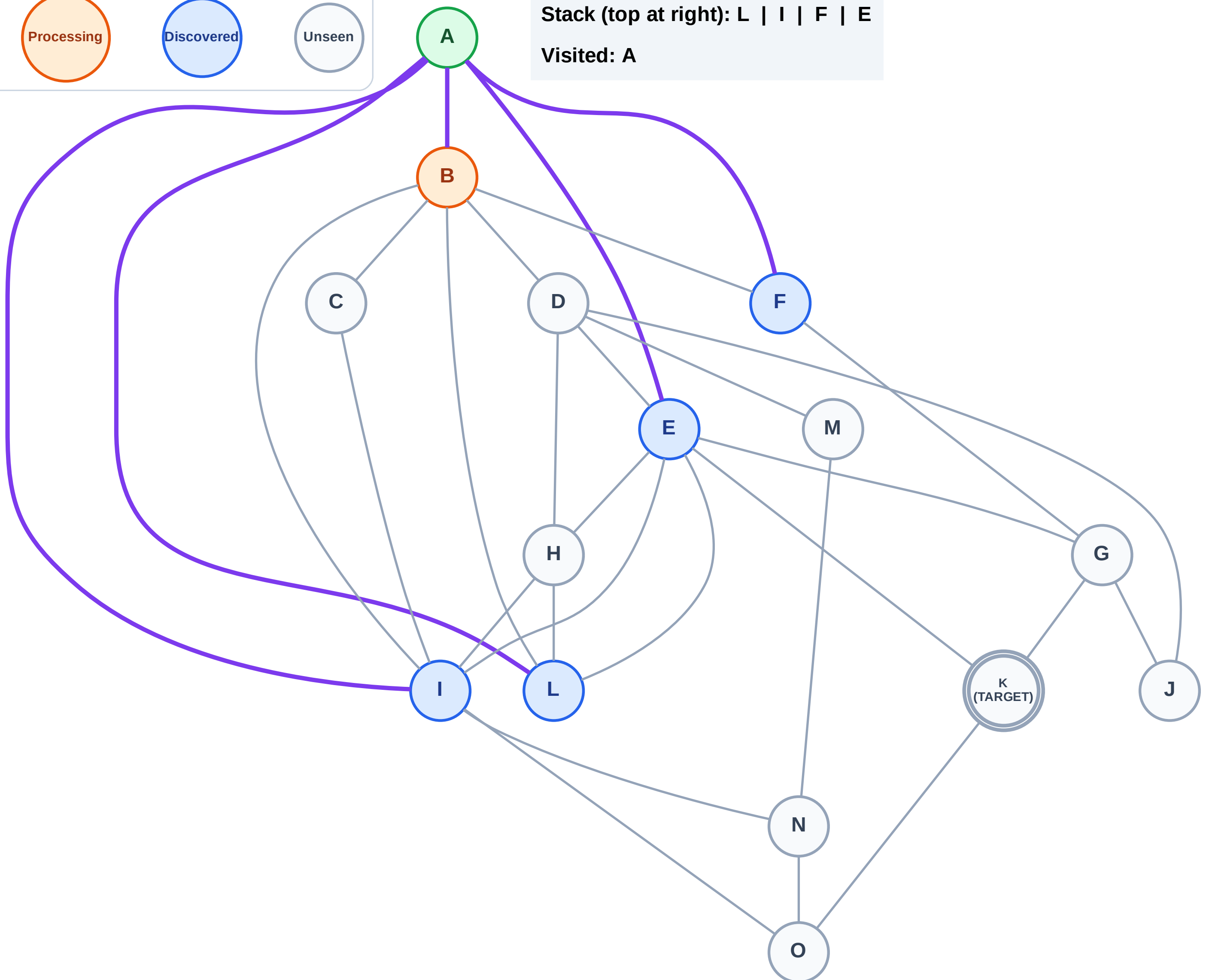
Step 4: Process node B

Legend



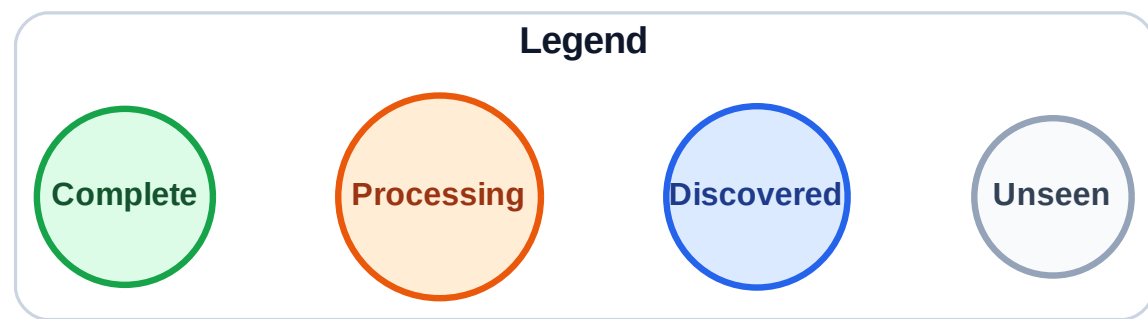
Stack (top at right): L | I | F | E

Visited: A



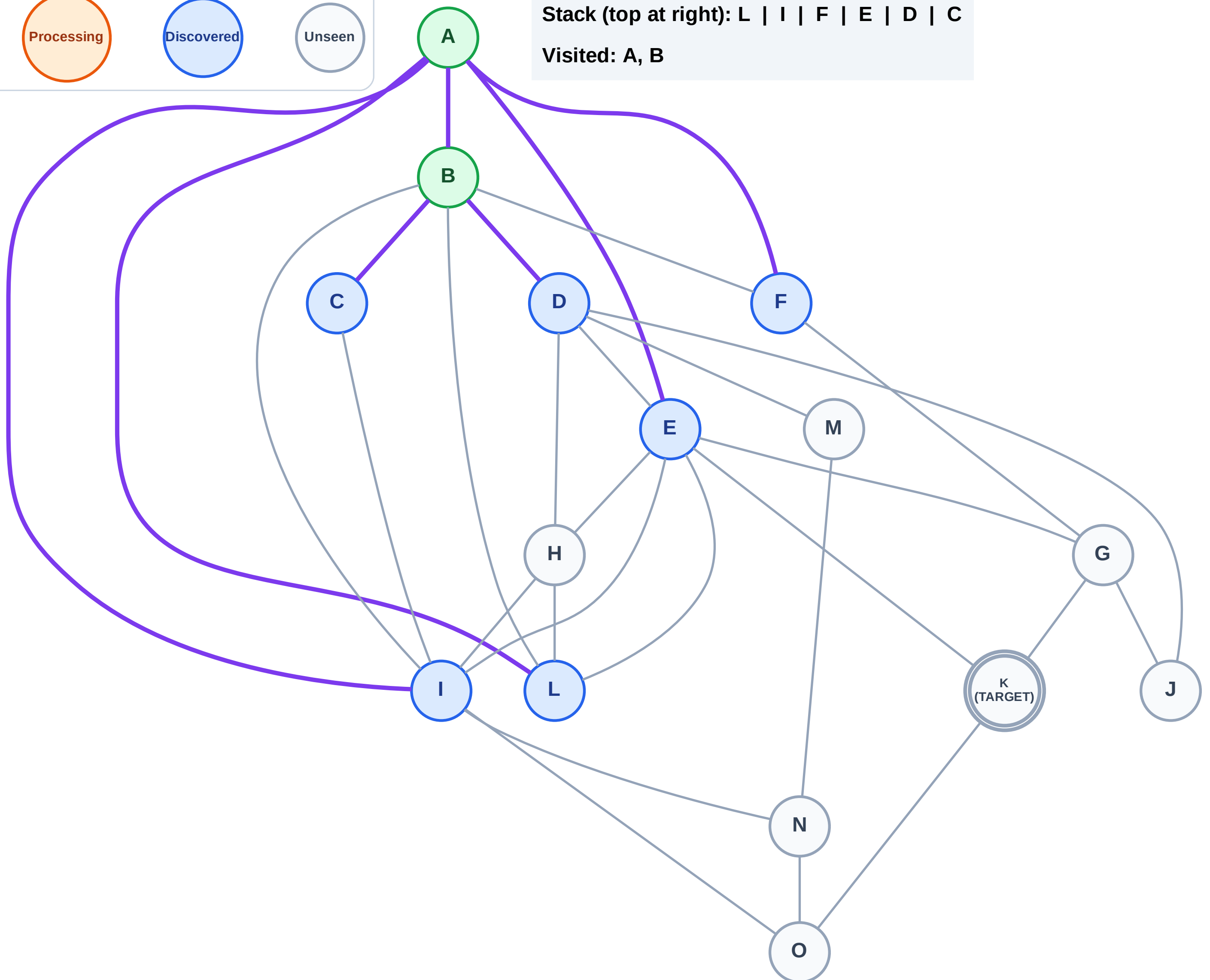
DFS Traversal

Step 5: Discover D, C from B



Stack (top at right): L | I | F | E | D | C

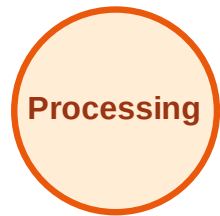
Visited: A, B



DFS Traversal

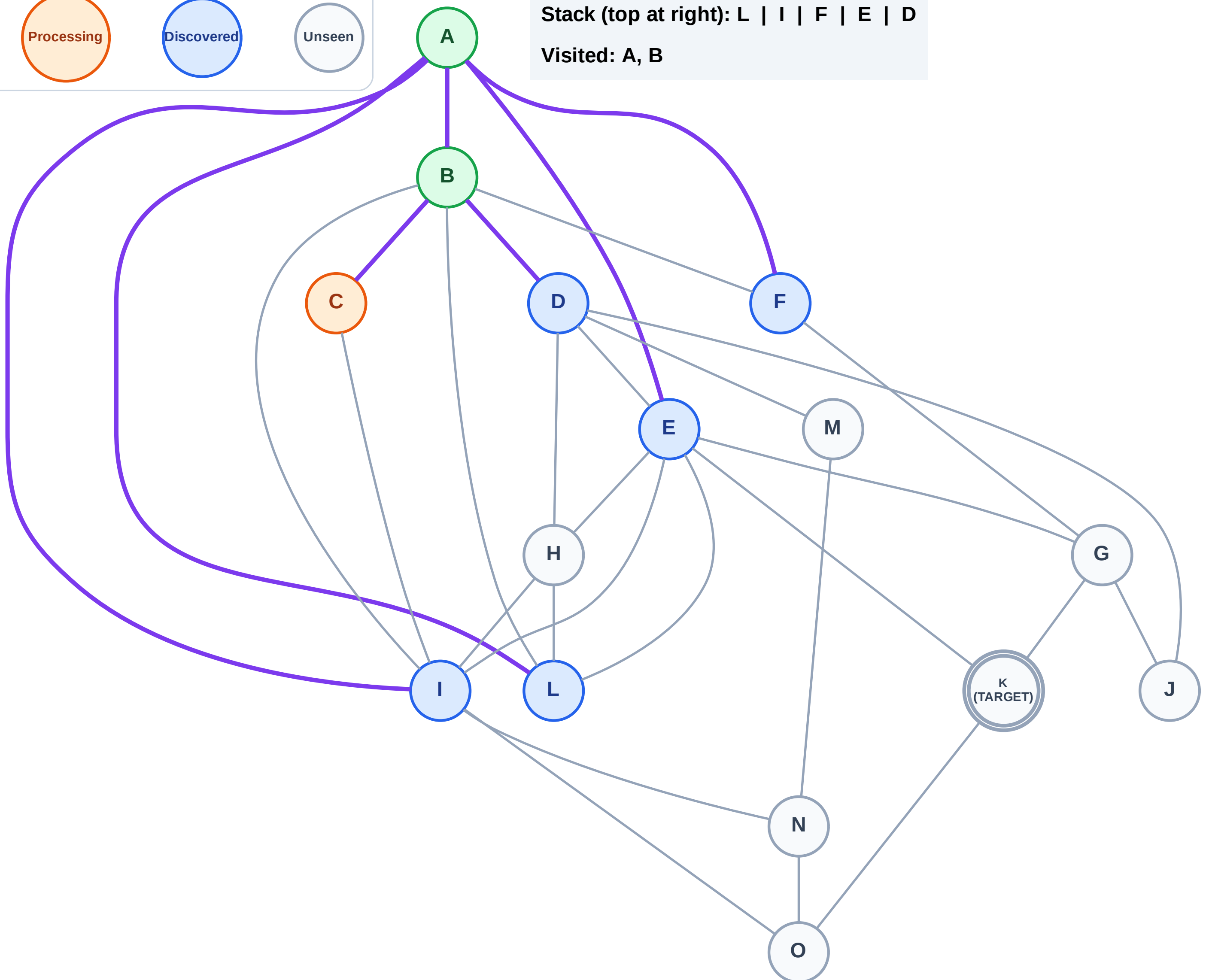
Step 6: Process node C

Legend



Stack (top at right): L | I | F | E | D

Visited: A, B



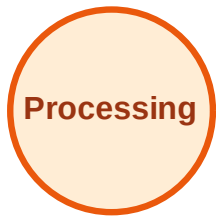
DFS Traversal

Step 7: No new neighbors from C

Legend



Complete



Processing



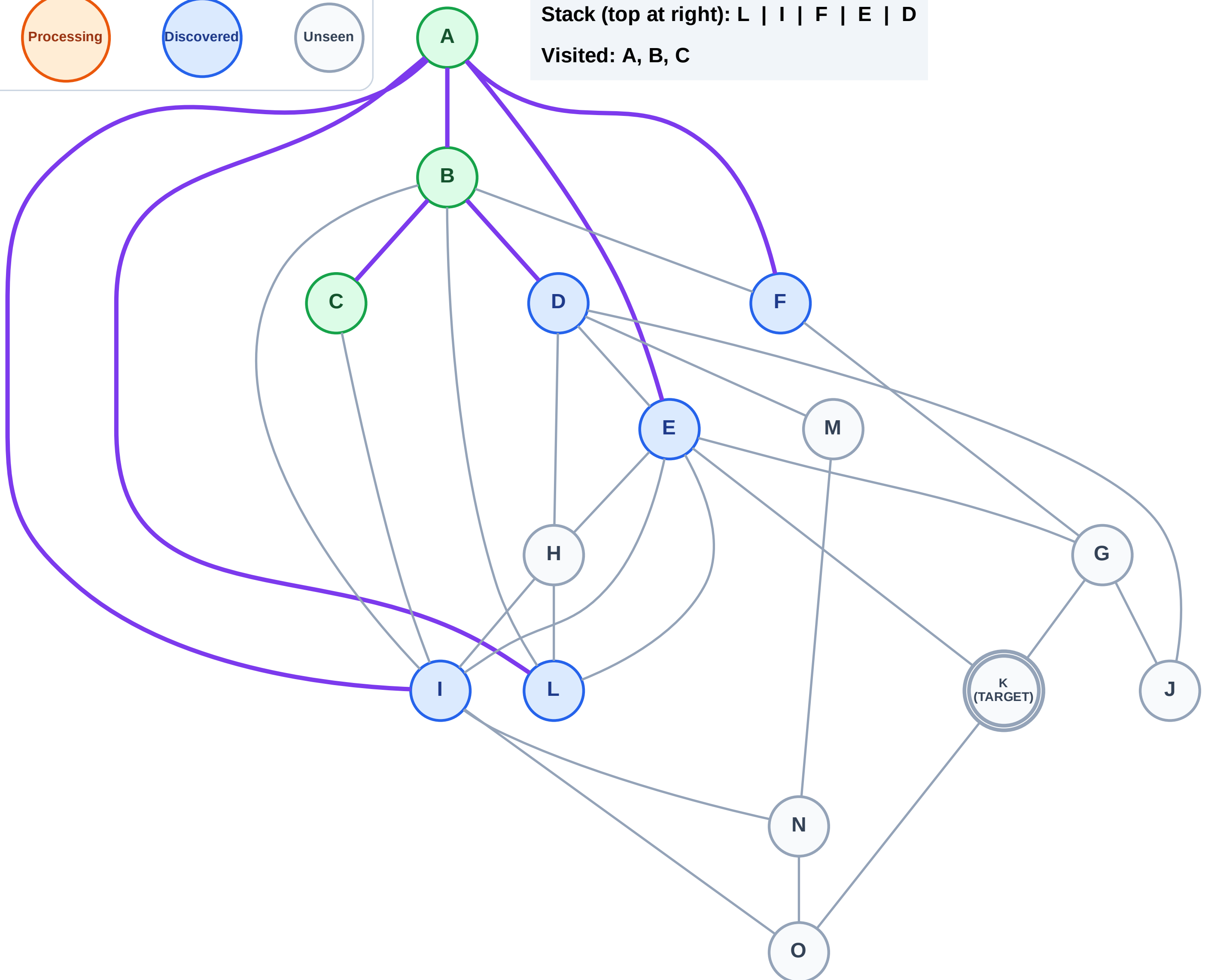
Discovered



Unseen

Stack (top at right): L | I | F | E | D

Visited: A, B, C



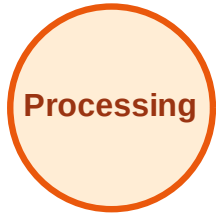
DFS Traversal

Step 8: Process node D

Legend



Complete



Processing



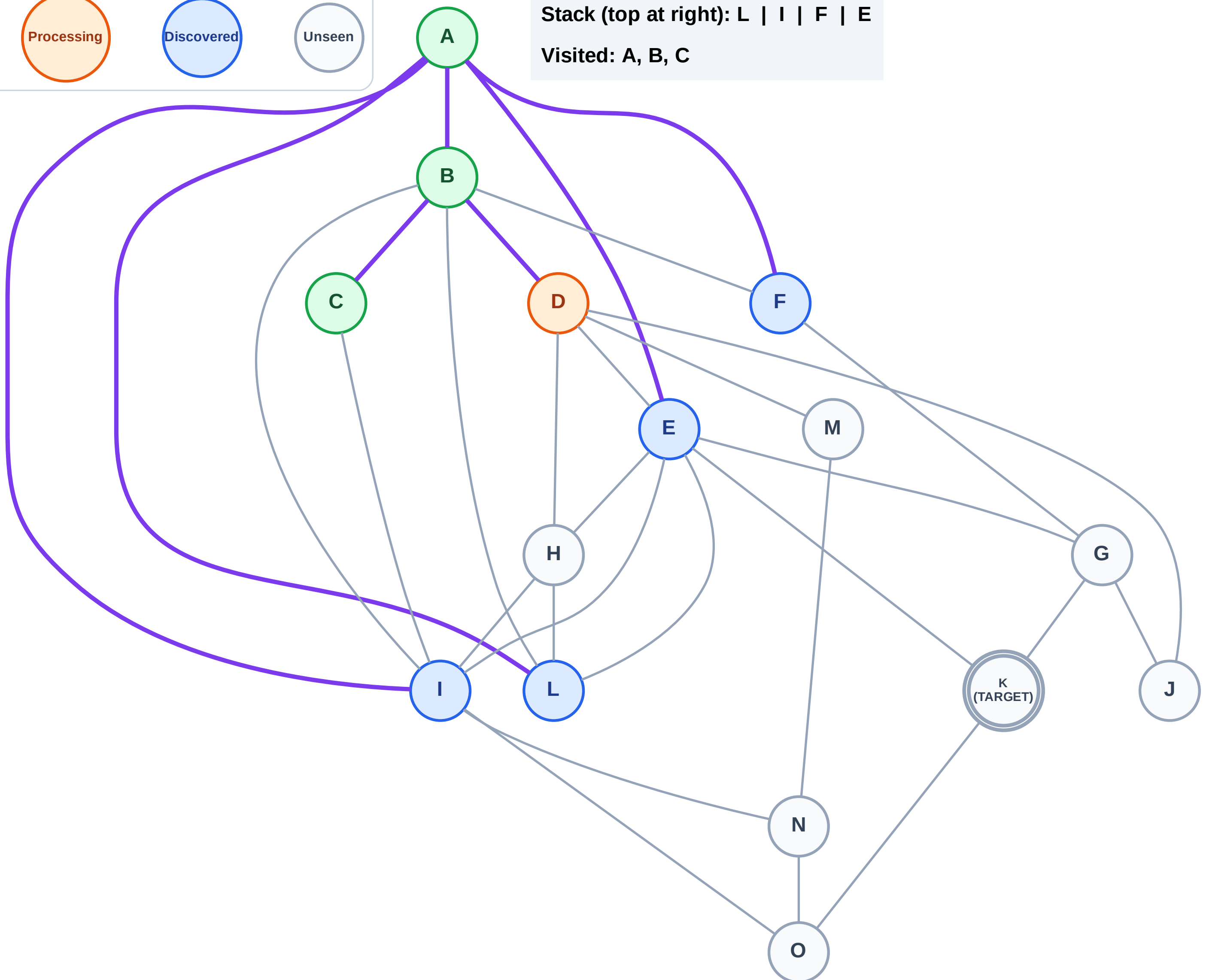
Discovered



Unseen

Stack (top at right): L | I | F | E

Visited: A, B, C

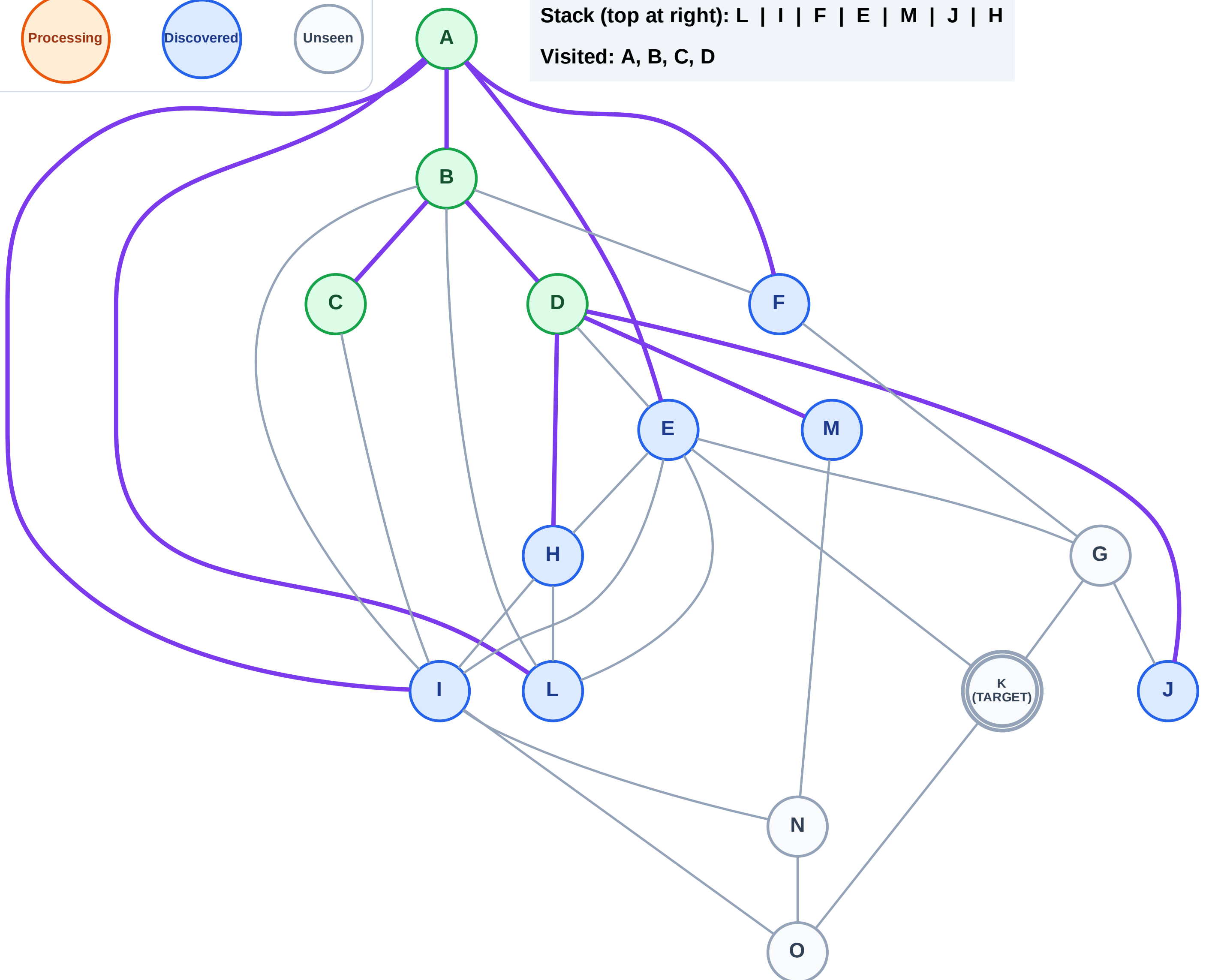


Step 9: Discover M, J, H from D

Stack (top at right): L | I | F | E | M | J | H

Legend

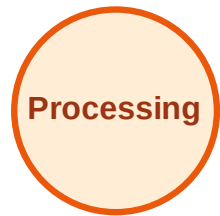
Unseen



DFS Traversal

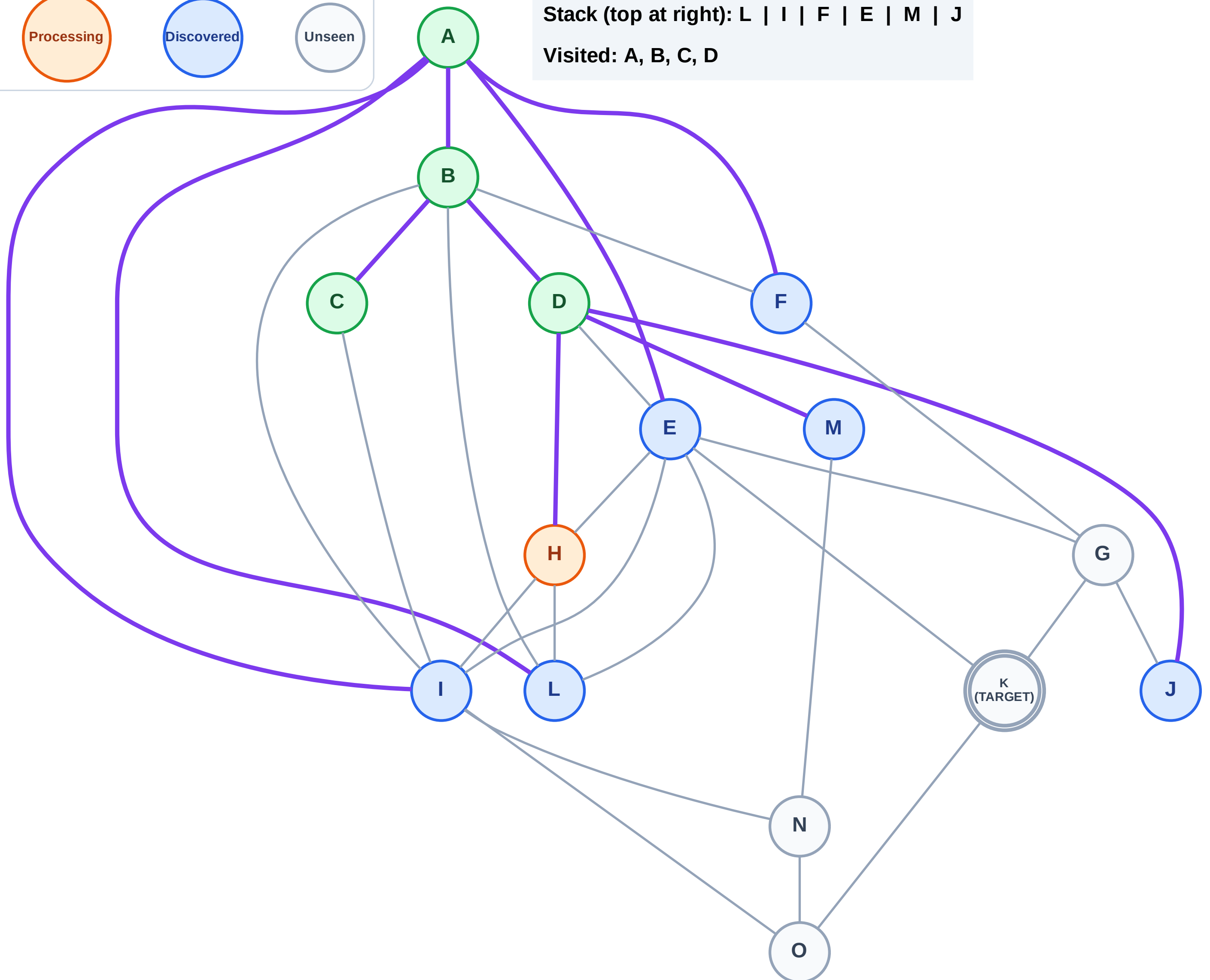
Step 10: Process node H

Legend



Stack (top at right): L | I | F | E | M | J

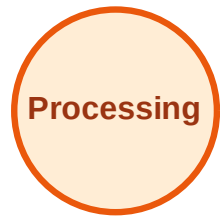
Visited: A, B, C, D



DFS Traversal

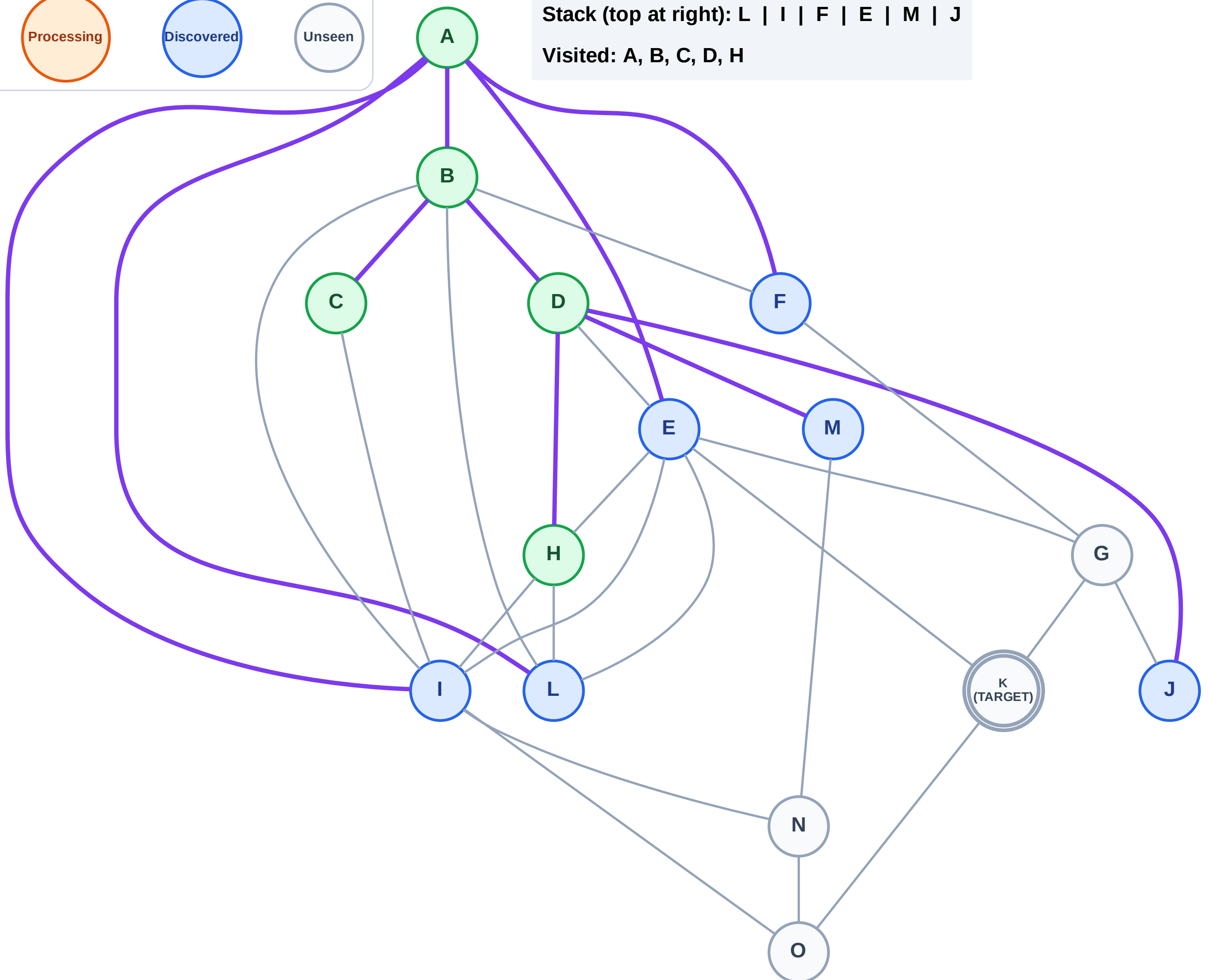
Step 11: No new neighbors from H

Legend



Stack (top at right): L | I | F | E | M | J

Visited: A, B, C, D, H



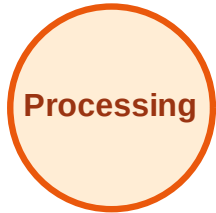
DFS Traversal

Step 12: Process node J

Legend



Complete



Processing



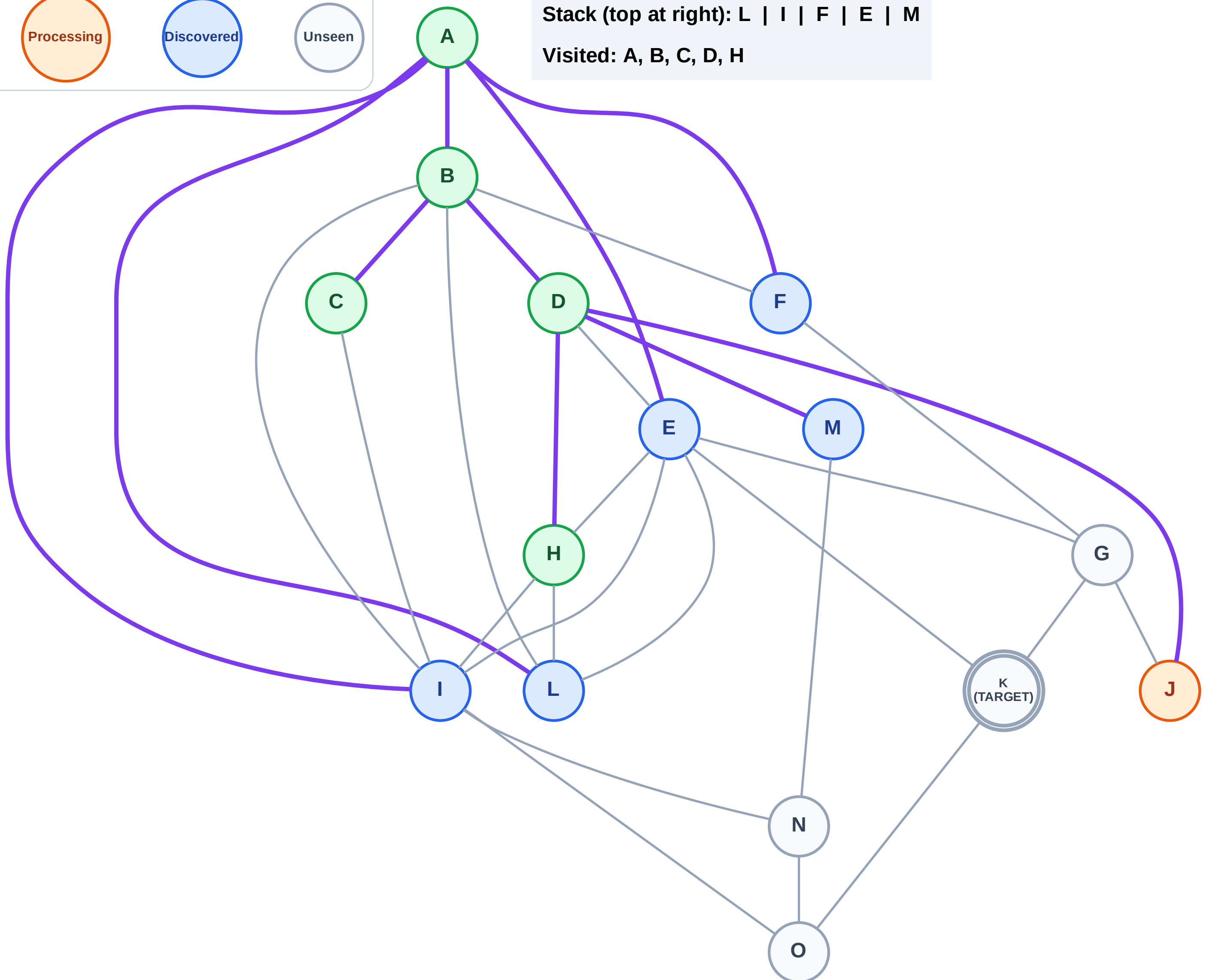
Discovered



Unseen

Stack (top at right): L | I | F | E | M

Visited: A, B, C, D, H



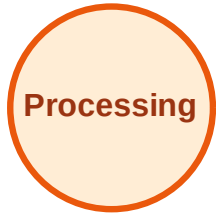
DFS Traversal

Step 13: Discover G from J

Legend



Complete



Processing



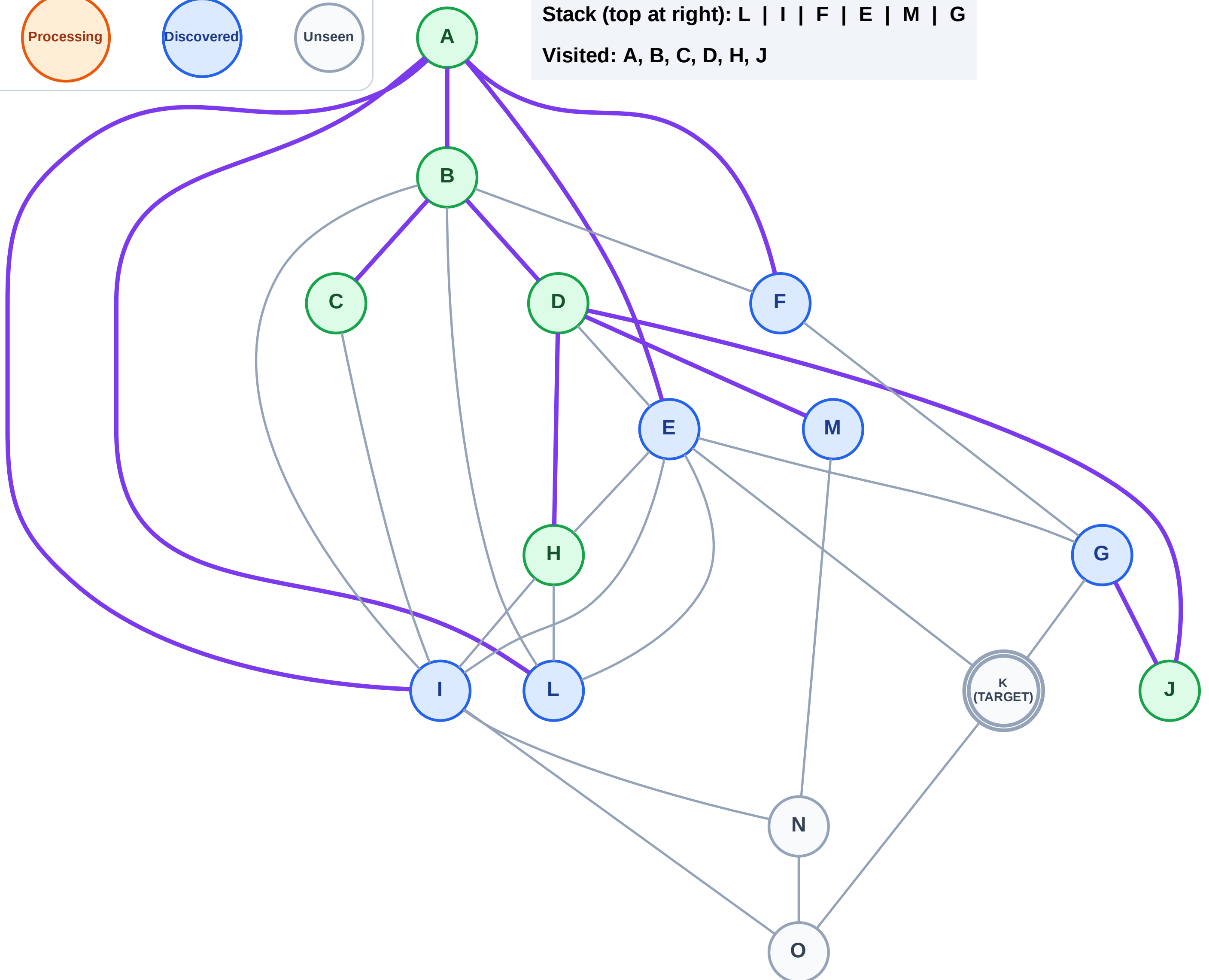
Discovered



Unseen

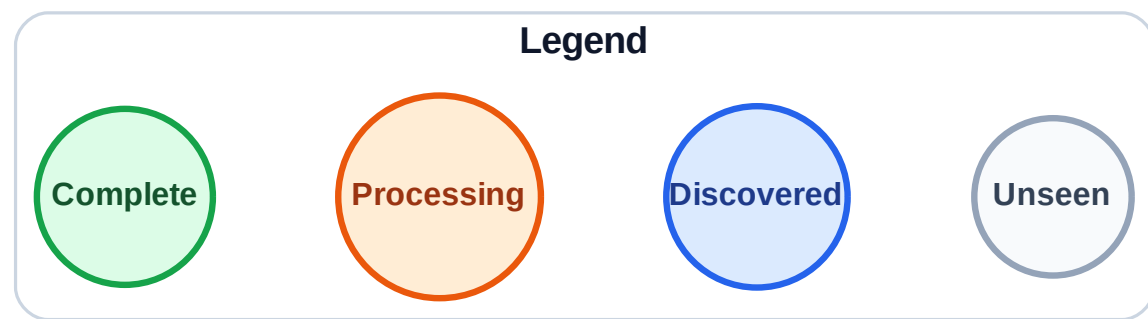
Stack (top at right): L | I | F | E | M | G

Visited: A, B, C, D, H, J

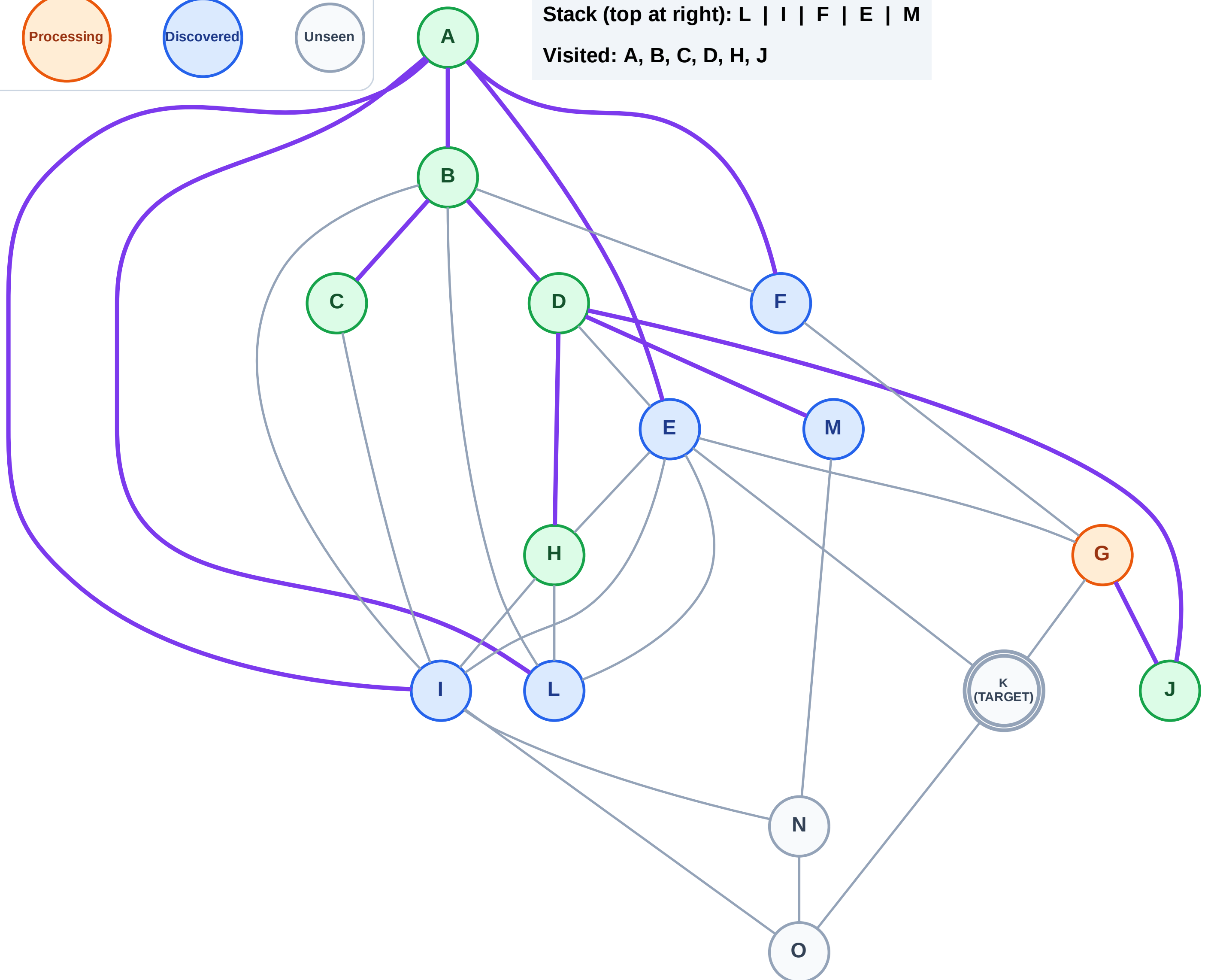


DFS Traversal

Step 14: Process node G

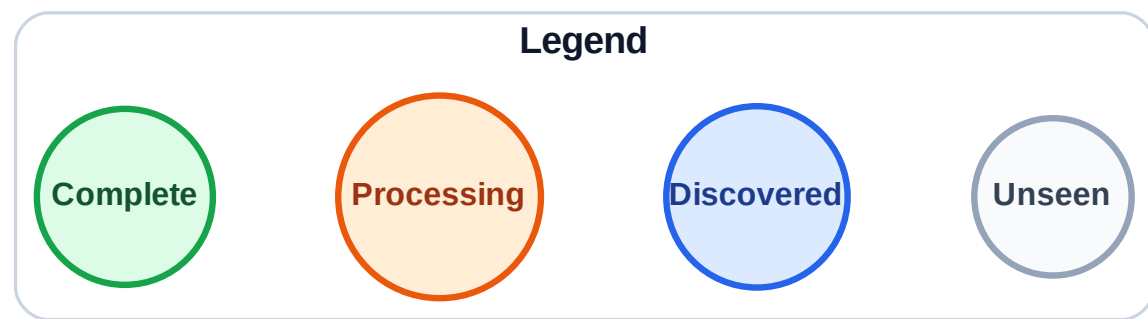


Stack (top at right): L | I | F | E | M
Visited: A, B, C, D, H, J

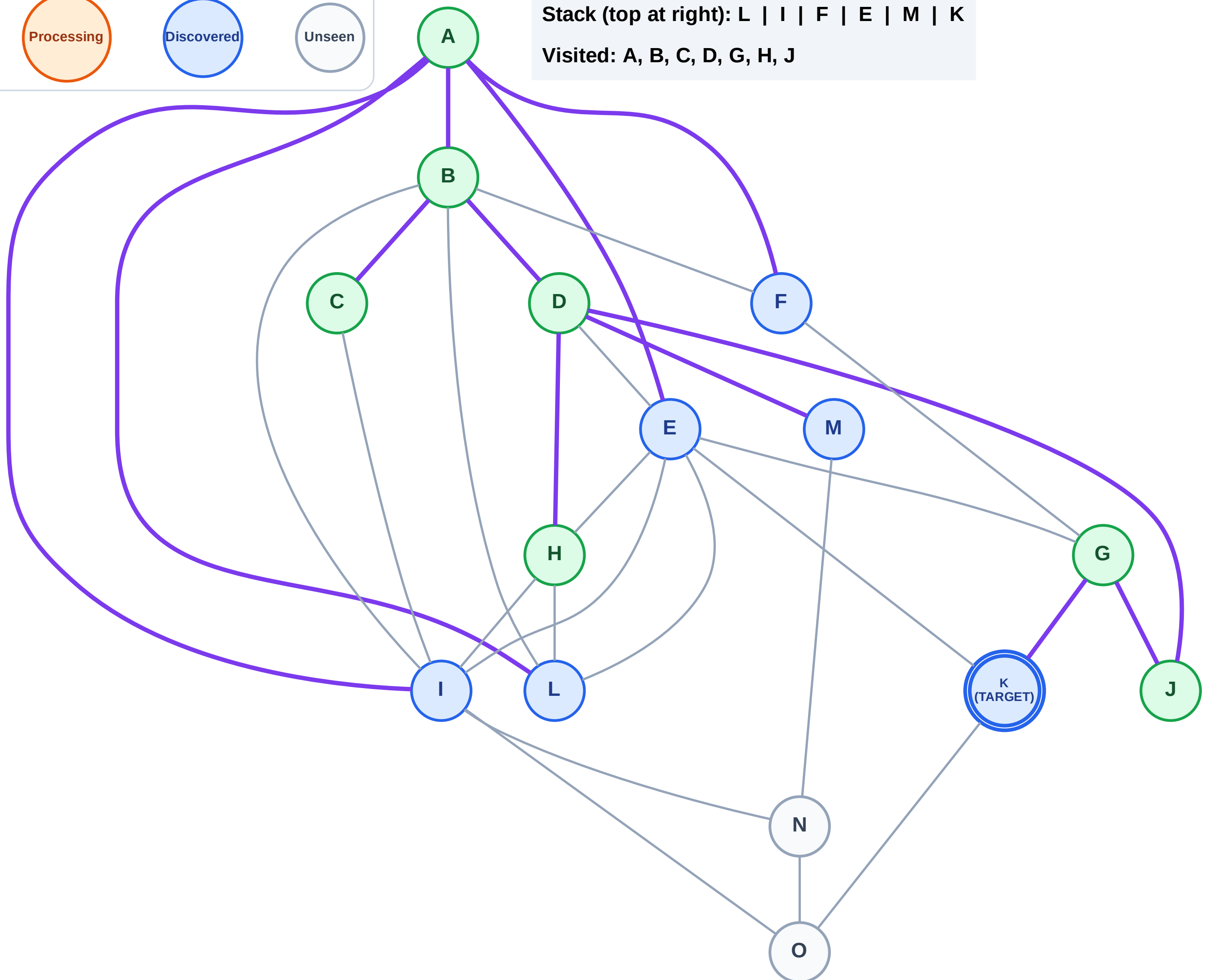


DFS Traversal

Step 15: Discover K from G



Stack (top at right): L | I | F | E | M | K
Visited: A, B, C, D, G, H, J



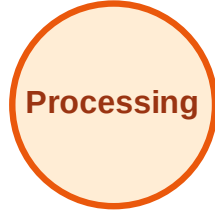
DFS Traversal

Step 16: Process node K

Legend



Complete



Processing



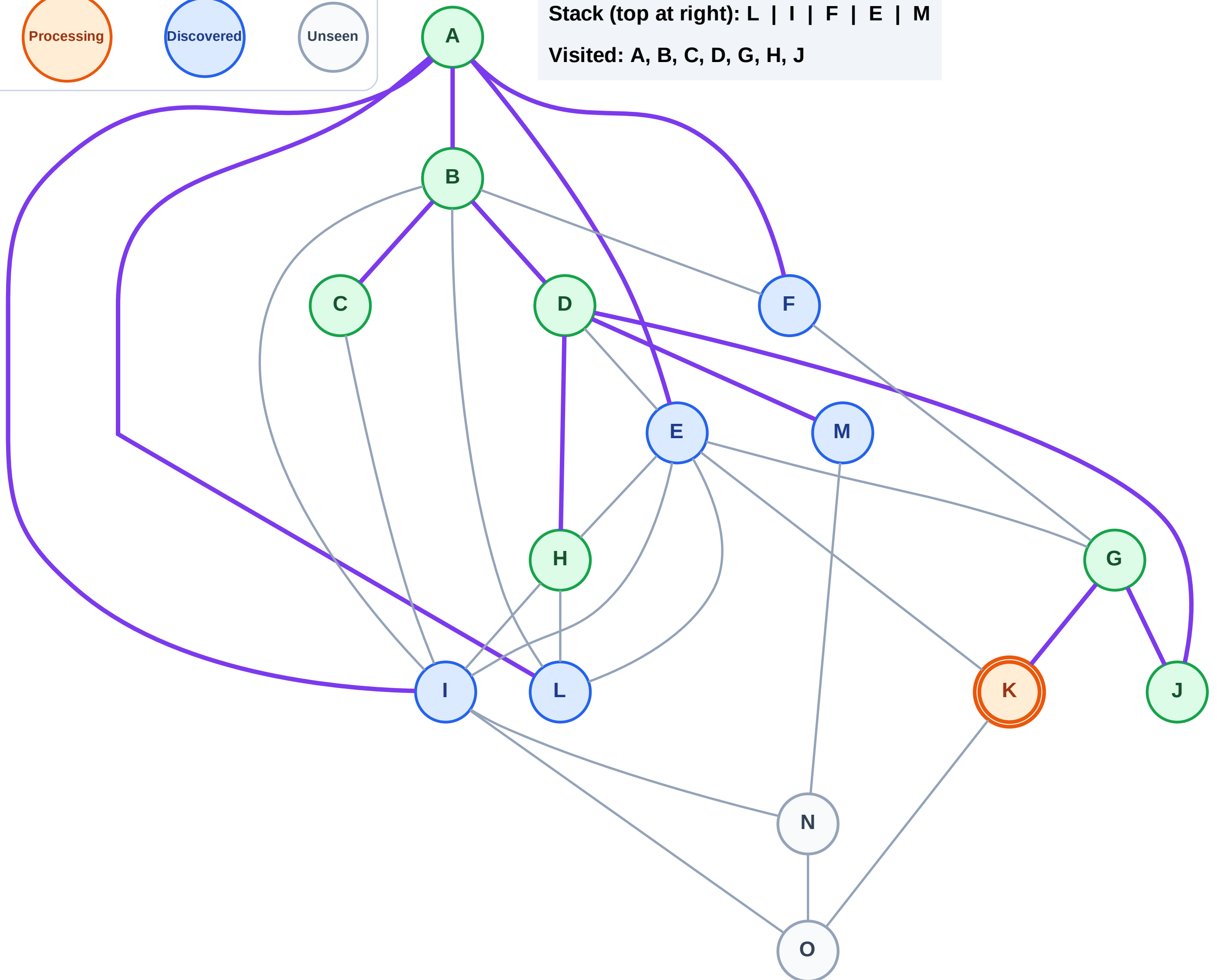
Discovered



Unseen

Stack (top at right): L | I | F | E | M

Visited: A, B, C, D, G, H, J



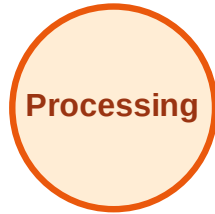
DFS Traversal

Step 17: TARGET NODE K REACHED! Path (5 hops): A → B → D → J → G → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): L | I | F | E | M

Visited: A, B, C, D, G, H, J, K

